PENALTY LEARNING FOR OPTIMAL PARTITIONING VIA INTERVAL REGRESSION

By

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ABSTRACT

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Changepoint detection identifies points in a sequence where the underlying data distribution changes, and it is widely used in fields such as finance, genomics, and healthcare. Dynamic programming algorithms, particularly those in the Optimal Partitioning (OP) family, are commonly used to detect changepoints efficiently. These methods rely on a penalty parameter to control the number of detected changepoints, but choosing an appropriate penalty is challenging: too low a penalty can lead to over-segmentation, while too high a penalty can miss important changes. Traditionally, penalty selection has been based on manually computed features of the sequence and simple predictive models, such as linear or tree-based regression. However, these approaches often fail to capture the complex, non-linear relationship between sequence characteristics and the optimal penalty. Furthermore, the optimal penalty is not a single value but lies within a range, which naturally motivates modeling the problem as interval regression. This thesis explores interval regression as a framework for learning penalty values in changepoint detection. I review existing interval regression methods and compare their performance on both synthetic and real-world datasets. Building on this, I propose learning-based approaches for predicting penalties directly from data, including multilayer perceptrons (MLPs) for continuous prediction and recurrent neural networks (RNNs) for sequences with temporal dependencies. These models reduce the need for manual feature engineering and better capture non-linear patterns in the data. Experiments on benchmark genomic datasets show that interval regression–based penalty learning improves changepoint detection performance, achieving higher accuracy and F1 scores than traditional approaches. The results also indicate that no single model works best in all cases, highlighting the importance of selecting a model suited to the characteristics of the data. Overall, this work demonstrates that interval regression provides a natural and effective approach to learning penalties for changepoint detection.

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PREFACE

This thesis is the result of several years of study and research on changepoint detection. The work presented here reflects my interest in combining statistical methods with machine learning to address practical challenges in data analysis. I would like to express my sincere gratitude to my primary supervisor Toby Dylan Hocking, and the research community for their guidance, support, and constructive feedback throughout this journey. While this dissertation focuses on technical contributions, it is also a personal exploration of the process of learning, experimentation, and discovery that comes with pursuing research.

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Dissertation

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1 Literature Review: Interval Regression

Definition In traditional supervised regression tasks, the objective is to develop a model that accurately predicts a continuous real-valued outcome based on a set of input features. This classical problem has been well-studied, resulting in a wide array of algorithms. However, a related but less explored problem is *interval regression*, where each instance is associated with an interval $y = [y_l, y_u]$ – lower and upper – that represents a range of acceptable predictions, rather than a single target value. Unlike conventional regression, where the goal is to predict a specific point, interval regression acknowledges the uncertainty or variability in the target, which is common in various real-world applications, such as Medical Research [Fazil Baksh et al., 2006, Rigaill et al., 2013, Hawkins, 2004], Economic Forecasting [Abbara and Zevallos, 2018], Environmental Studies [Ellison et al., 2014], Engineering [Levitin et al., 2011]. Refer to Figure 1 for a visualization of the interval regression problem setting.

The outcome data may be uncensored ($-\infty < y_l = y_u < \infty$), right-censored ($-\infty < y_l < y_u = \infty$), left-censored ($-\infty = y_l < y_u < \infty$), or interval-censored ($-\infty < y_l < y_u < \infty$) depending on the available information.

Literature Review Interval-valued outputs commonly occur in survival analysis area. In some studies, Interval Regression models are trained by converting the problem into Standard Regression: instead of using intervals directly as targets, the target interval is represented by discrete points, which are then treated as Standard Regression targets. Using the same notations in Table 1

	training dataset			
	features	target	prediction	this study
Standard Regression	$\mathbf{x} \in \mathbf{R}^m$	$y \in \mathbf{R}$	$\hat{y} \in \mathbf{R}$	✗
Interval Regression	$\mathbf{x} \in \mathbf{R}^m$	$(y_l, y_u) \in \mathbf{R}^2$	$\hat{y} \in \mathbf{R^2}$	✗
	$(\mathbf{x}_l, \mathbf{x}_u) \in \mathbf{R}^{m\times 2}$			✗
	$\mathbf{x} \in \mathbf{R}^m$	$(y_l, y_u) \in \mathbf{R}^2$	$\hat{y} \in \mathbf{R}$	✓

Table 1: Comparison of Standard Regression versus Interval Regression setting.

for $\mathbf{x}$ as the features and (y_l, y_u) as the target interval, Drouin et al. [2017] transform the instance $\big(\mathbf{x}, (y_l, y_u)\big)$ into two points, $(\mathbf{x}, y_l)$ and $(\mathbf{x}, y_u)$, called Interval-CART model – Approach 1 in Figure 2, whereas Cheng et al. [2023] convert it into its midpoint, $(\mathbf{x}, \frac{y_l+y_u}{2})$ – Approach 2 in Figure 2. While these approaches seem straightforward and reasonable, it overlooks the interval information by representing intervals by finite values and fails to handle left-censored or right-censored intervals, resulting in wasted data. Consequently, the performance of these models is worse compared to the models specifically designed for interval targets, see the results in [Drouin et al., 2017, Cheng et al., 2023] and the example in Figure 2.

Linear models for censored outcomes, known as *accelerated failure time* (AFT) models, have been thoroughly explored [Wei, 1992]. More recently, variants with L1 regularization have emerged to handle high-dimensional datasets [Cai et al., 2009, Huang et al., 2006]. Nonlinear approaches to censored data have also been investigated, including Support Vector Machines [Pölsterl et al., 2016] and decision tree-based models such as Random Survival Trees [Ishwaran et al., 2008], Cox-Boosting [Binder and Schumacher, 2008], Bagging Survival Trees [Hothorn et al., 2004], Scikit-Survival [Pölsterl, 2020] and Cox-PH in XGBoost [Lundberg et al., 2018]. However, most of these models are limited to right-censored data. Recently, the AFT model in XGBoost [Barnwal et al., 2022] has emerged as a tree-based model that claims to support all types of label censoring. However, this AFT model does not inherently support negative outcomes, so they addressed this by applying an exponential transformation to the output, ensuring non-negative interval-censored labels. In addition to AFT models, interval regression can also be approached through supervised machine learning techniques that use gradient descent optimizers to minimize a specific loss function (different from negative log-likelihood). This includes L1-regularized linear models with squared hinge loss [Rigaill et al., 2013] and non-linear models such as maximal margin interval trees (MMIT) [Drouin et al., 2017]. The latter generalizes the widely known Classification and Regression Tree (CART) algorithm [Breiman et al., 1984] to accommodate censored outputs.

This following sub-sections provides a technical overview of the interval regression method, consisting of two main subsections: AFT models (a statistical approach) and supervised machine

model name	left-censored	right-censored	interval-censored	scalar output	base learner	citation
standard linear	✗	✗	✗	✓	Linear	[Legendre, 1805]
standard knn	✗	✗	✗	✓	kNN	[Fix and Hodges, 1985]
standard tree	✗	✗	✗	✓	Tree	[Breiman et al., 1984]
standard forest	✗	✗	✗	✓	Ensemble trees	[Breiman, 2001]
standard mlp	✗	✗	✗	✓	MLP	[Hornik et al., 1989]
interval-linear	✓	✓	✓	✗	Linear	[Wang et al., 2015]
quadratic	✓	✓	✓	✗	Linear	[Tanaka and Lee, 1998]
svm	✓	✓	✓	✗	Linear	[Jeng et al., 2003]
aft	✗	✓	✗	✓	Linear	[Wei, 1992]
reg-aft	✗	✓	✗	✓	Linear	[Huang et al., 2006]
middlemost	✗	✗	✓	✓	Linear, MLP	[Cheng et al., 2023]
interval-CART	✗	✗	✓	✓	Tree	[Drouin et al., 2017]
interval-knn	✓	✓	✓	✗	kNN	[Hamed et al., 2012]
linear	✓	✓	✓	✓	Linear	[Rigaill et al., 2013]
mmit	✓	✓	✓	✓	Tree	[Drouin et al., 2017]
aft_xgb	✓	✓	✓	✓	Ensemble trees	[Barnwal et al., 2022]
constant	✓	✓	✓	✓	Constant	baseline
knn	✓	✓	✓	✓	kNN	proposed
mlp	✓	✓	✓	✓	MLP	proposed
mmif	✓	✓	✓	✓	Ensemble trees	proposed

Table 2: List of Regression models with compatible censored target intervals.

learning models. In essence, AFT models assume the data follows a specific distribution and aim to maximize the corresponding likelihood, while supervised machine learning models focus on minimizing their respective loss functions.

1.1 AFT Models

1.1.1 Original AFT Model

The Accelerated Failure Time (AFT) model is a parametric survival model that directly models the time to an event. The model assumes that covariates act multiplicatively on the survival time, effectively accelerating or decelerating the event. The basic AFT model can be expressed as:

$$\log(T_i) = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i$$

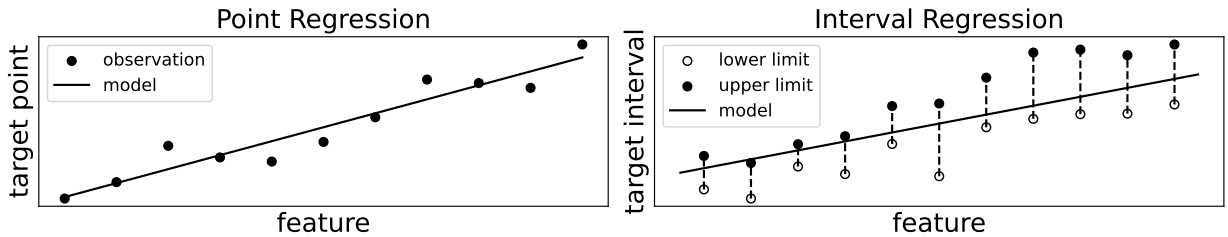


Figure 1: Example of point regression versus interval regression

where:

- T_i is the survival time (non-negative) for the i-th individual,
- $\mathbf{x}_i$ is the vector of covariates for the i-th individual,
- $\boldsymbol{\beta}$ is the vector of regression coefficients,
- ϵ_i is a random error term following a specified distribution.

Censored Data and Likelihood Function In survival analysis, the data may be *right-censored*, meaning the event of interest has not occurred by the end of the study. The likelihood function must account for both observed and censored data.

Let δ_i be an indicator variable, where:

$$\delta_i = \begin{cases} 1 & \text{if the event is observed,} \\ 0 & \text{if the observation is censored.} \end{cases}$$

The likelihood function for the AFT model with censored data is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \left[f(T_i \mid \mathbf{x}_i, \boldsymbol{\beta})\right]^{\delta_i} \left[S(T_i \mid \mathbf{x}_i, \boldsymbol{\beta})\right]^{1-\delta_i}$$

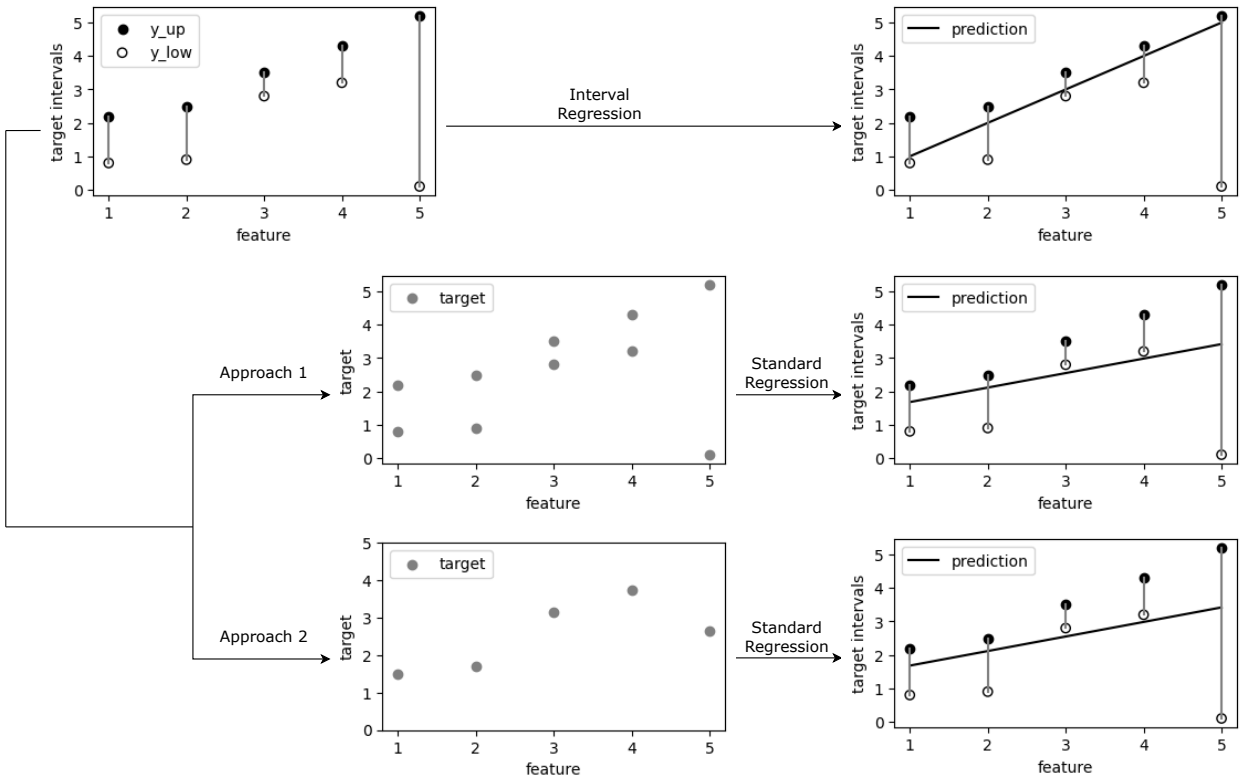


Figure 2: Example of converting Interval Regression into Standard Regression

where $f(T_i \mid \mathbf{x}_i, \boldsymbol{\beta})$ is the probability density function, and $S(T_i \mid \mathbf{x}_i, \boldsymbol{\beta}) = P(T_i > t)$ is the survival function. The survival function represents the probability that the event has not occurred by a certain time t, i.e.,

$$S(T_i \mid \mathbf{x}_i, \boldsymbol{\beta}) = 1 - F(T_i \mid \mathbf{x}_i, \boldsymbol{\beta}),$$

where $F(T_i \mid \mathbf{x}_i, \boldsymbol{\beta})$ is the cumulative distribution function (CDF) of the event time T_i conditional on the covariates $\mathbf{x}_i$ and model parameters $\boldsymbol{\beta}$.

Taking the logarithm of the likelihood function yields the log-likelihood:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[\delta_i \log f(T_i \mid \mathbf{x}_i, \boldsymbol{\beta}) + (1 - \delta_i) \log S(T_i \mid \mathbf{x}_i, \boldsymbol{\beta})\right]$$

Model Fitting Procedure To estimate the parameters $\boldsymbol{\beta}$, we maximize the log-likelihood function. The general steps for training the AFT model are:

1. Specify the distribution for the error term ϵ_i (e.g., normal, log-normal, Weibull).
2. Define the likelihood function based on the observed and censored data.
3. Maximize the log-likelihood function to estimate $\boldsymbol{\beta}$.
4. Use the estimated parameters to make predictions.

Example: Survival Analysis with Normal Distribution Let us consider the case where the survival times follow a normal distribution, i.e., $\epsilon_i \sim N(0, \sigma^2)$. The model is given by:

$$\log(T_i) = \beta_0 + \beta_1 x_{i1} + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

The corresponding log-likelihood function for this model is:

$$\ell(\boldsymbol{\beta}, \sigma) = \sum_{i=1}^{n} \left[\delta_i \log \left(\frac{1}{\sigma\sqrt{2\pi}} \exp \left(-\frac{(\log T_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\sigma^2} \right) \right) + (1 - \delta_i) \log S(T_i \mid \mathbf{x}_i, \boldsymbol{\beta}, \sigma) \right]$$

where $S(T_i \mid \mathbf{x}_i, \boldsymbol{\beta}, \sigma)$ is the survival function for the normal distribution:

$$S(T_i \mid \mathbf{x}_i, \boldsymbol{\beta}, \sigma) = 1 - \Phi \left(\frac{\log T_i - \mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma} \right)$$

and $\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution.

The parameters β_0, β_1, and σ are estimated by maximizing the log-likelihood function. Predictions for new individuals can be made by:

$$\hat{T}_{\text{new}} = \exp(\mathbf{x}_{\text{new}}^\top \hat{\boldsymbol{\beta}} + \hat{\epsilon})$$

model	source code	language	citation
linear	[Hocking, 2024]	R	[Rigaill et al., 2013]
mmit	[Drouin et al., 2017]	C++	[Drouin et al., 2017]
aft_xgb	[Chen and Guestrin, 2016]	Python	[Barnwal et al., 2022]
constant	paper GitHub repo	C++	additional baseline
knn	paper GitHub repo	Python	proposed
mlp	paper GitHub repo	Python	proposed
mmif	paper GitHub repo	C++	proposed

Table 3: List of the regression models in this with corresponding source.

where $\hat{\epsilon}$ is drawn from $N(0, \hat{\sigma}^2)$.

1.1.2 AFT Model in XGBoost

The AFT model in XGBoost, introduced by Barnwal et al. [2022], is an ensemble tree-based model designed for survival analysis. The model makes predictions as follows:

$$y_p = \mathbf{T}(\mathbf{x}),$$

where $\mathbf{T}$ represents the ensemble of trees in XGBoost. Although it belongs to the AFT model family, this model can handle all non-negative target intervals. To extend its applicability to real-valued target intervals $(y_l, y_u) \in \mathbf{R}^2$, Barnwal et al. [2022] apply an exponential transformation to the original interval:

$$(y_l, y_u) \longrightarrow (\exp y_l, \exp y_u) \in \mathbf{R}_+^2$$

After making a prediction y_p in the exponential space, the result is mapped back to the original scale using the logarithm transformation:

$$\hat{y} = \log y_p = \log \mathbf{T}(\mathbf{x})$$

Training this model is based on minimizing the negative log-likelihood function (loss function),

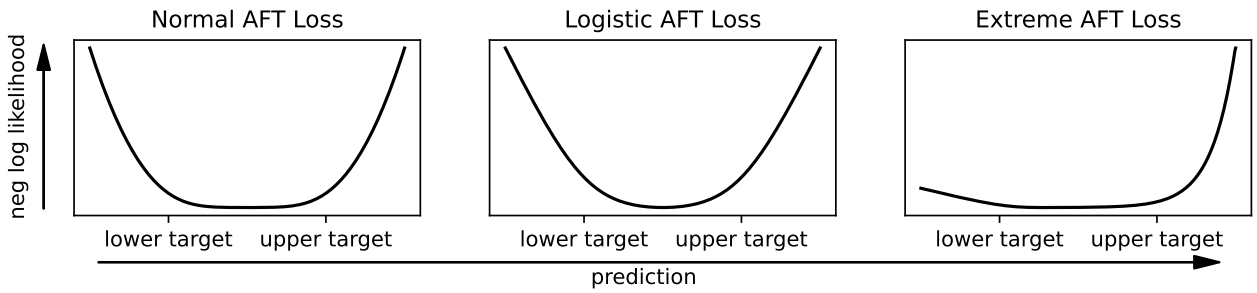


Figure 3: Visualization of the AFT XGBoost Loss (negative log-likelihood) for 3 distributions

which incorporates all training survival instances:

$$l_{AFT}\Big(\hat{y}, (y_l, y_u)\Big) = -\log\Big[F_Z\big(\frac{y_u - \hat{y}}{\sigma}\big) - F_Z\big(\frac{y_l - \hat{y}}{\sigma}\big)\Big] \tag{1}$$

where $\sigma > 0$ is the scale parameter. The cumulative distribution function $F_Z(z)$ depends on the chosen distribution (normal, logistic, or extreme):

- normal: $F_Z(z) = \frac{1}{2}\big(1 + \text{erf}(\frac{z}{\sqrt{2}})\big)$
- logistic: $F_Z(z) = \frac{e^z}{1+e^z}$
- extreme: $F_Z(z) = 1 - e^{-\exp z}$

Figure 3 visualizes the loss function for each distributions, which can be compared to Figure 4 to observe their similarities. The model is trained using the gradient boosting algorithm [Friedman, 2001] to minimize the loss value (8) over the training dataset with N instances $\{\mathbf{x}^i, (y_l^i, y_u^i)\}_{i=1}^N$ given the distribution and the scale parameter σ:

$$\sum_{i=1}^{N} l_{AFT}\Big(\hat{y}^i, (y_l^i, y_u^i)\Big) = \sum_{i=1}^{N} -\log\Big[F_Z\big(\frac{y_u^i - \log \mathbf{T}(\mathbf{x}^i)}{\sigma}\big) - F_Z\big(\frac{y_l^i - \log \mathbf{T}(\mathbf{x}^i)}{\sigma}\big)\Big]$$

1.2 Supervised Machine Learning Models

Before delving into the methods, it is essential to clarify the loss function that these models utilized in interval regression. While the mean squared error (MSE) or mean absolute error (MAE) is com-

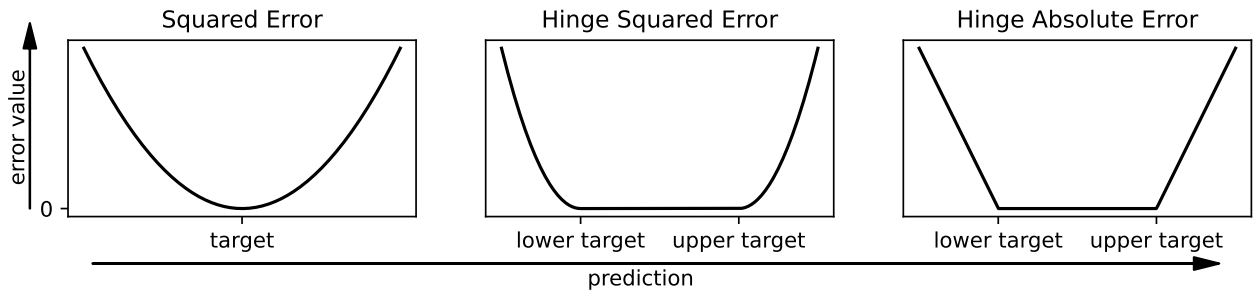


Figure 4: Visualization of the Loss Functions: Error values relative to predictions and targets

monly used for traditional regression tasks, interval regression employs a related criterion known as hinge error. The hinge loss function, leveraging the ReLU activation function, is defined as:

$$l(\hat{y}, y) = l(\hat{y}, [y_l, y_u]) = \big(\text{ReLU}(y_l - \hat{y} + \epsilon)\big)^p + \big(\text{ReLU}(\hat{y} - y_u + \epsilon)\big)^p \tag{2}$$

where $\epsilon \geq 0$ is the margin (or hinge), $\hat{y}$ denotes the predicted value and $y = [y_l, y_u]$ represents the target interval with $y_l \leq y_u$. p is either 1 or 2 ($p = 1$ is equivalent to MAE, $p = 2$ is equivalent to MSE). This formulation ensures that the loss is minimized when the predicted value $\hat{y}$ falls within the target interval $[y_l, y_u]$. Unlike MSE or MAE, which achieves zero loss at a single target value, the hinge error function yields zero loss within a specified target interval, as illustrated in Figure 4.

From this point forward in the study, the following notation will be used: $\mathbf{x}$ represents the feature vector, $[y_l, y_u]$ denotes the target interval, and $\hat{y}$ stands for the predicted output.

1.2.1 Linear

This linear model was introduced in the first time by Rigaill et al. [2013], and its extended theoretical foundations were later studied by Cheng et al. [2023]. The prediction formula is given:

$$\hat{y} = \mathbf{x} \cdot \beta + \beta_0 \tag{3}$$

where $\beta \in \mathbf{R}^m$ and $\beta_0 \in \mathbf{R}$ are the parameters to be learned. To estimate these parameters, the authors extended the Standard Regression loss function—specifically, the Squared Error or Mean

Squared Error (MSE)—into a generalized form called the *Hinge Squared Error*. The error value between prediction $\hat{y}$ and target interval (y_l, y_u) is defined as follows:

$$l\Big(\hat{y}, (y_l, y_u)\Big) = \begin{cases} (y_l - \hat{y} + \epsilon)^2, & \text{if } \hat{y} < y_l + \epsilon \\ 0, & \text{if } y_l + \epsilon \leq \hat{y} \leq y_u - \epsilon \\ (\hat{y} - y_u + \epsilon)^2, & \text{if } \hat{y} > y_u - \epsilon \end{cases}$$

where $\epsilon \geq 0$ is the margin length ($\epsilon = 0$ by default). The loss function becomes MSE when $y_l = y_u$ and $\epsilon = 0$, or in other word, target interval becomes target point.

Cheng et al. [2023] proposed a model called *Regression with Interval Targets* (RIT), which is essentially the same as model (11), but uses *Hinge Absolute Error* instead:

$$l\Big(\hat{y}, (y_l, y_u)\Big) = \begin{cases} |y_l - \hat{y} + \epsilon|, & \text{if } \hat{y} < y_l + \epsilon \\ 0, & \text{if } y_l + \epsilon \leq \hat{y} \leq y_u - \epsilon \\ |\hat{y} - y_u + \epsilon|, & \text{if } \hat{y} > y_u - \epsilon \end{cases}$$

The general Interval Regression loss function can be written in this short form:

$$l\Big(\hat{y}, (y_l, y_u)\Big) = \Big(\text{ReLU}(y_l - \hat{y} + \epsilon)\Big)^p + \Big(\text{ReLU}(\hat{y} - y_u + \epsilon)\Big)^p \tag{4}$$

where $p \in \{1, 2\}$. Refer to Figure 4 for the visual comparison between hinge errors (Interval Regression Loss) and squared error (Standard Regression Loss).

To estimate the parameters β and β_0 in (11), apply a gradient descent algorithm with respect to β and β_0 to minimize the Hinge Error (2) over the training dataset having N instances $\{\mathbf{x}^i, (y_l^i, y_u^i)\}_{i=1}^N$ given by:

$$\sum_{i=1}^{N} l\Big(\hat{y}^i, (y_l^i, y_u^i)\Big) = \sum_{i=1}^{N} l\Big(\mathbf{x^i} \cdot \beta + \beta_0, (y_l^i, y_u^i)\Big)$$

Rigaill et al. [2013] employed the Fast Iterative Shrinkage-Thresholding Algorithm or FISTA [Beck and Teboulle, 2009], while Cheng et al. [2023] employed stochastic gradient descent (SGD).

1.2.2 Tree

MMIT introduced by Drouin et al. [2017] is applicable for λ prediction as an interval regression model. The MMIT model is:

$$\log \hat{\lambda} = \mathcal{T}(\mathbf{x})$$

The tree $\mathcal{T}$ architecture follows the same structure as CART (Classification And Regression Tree) [Breiman et al., 1984]. The only difference lies in the regression value for each leaf which is a set of Q instances $\{\mathbf{x}^i, [\log \lambda_l^i, \log \lambda_u^i]\}_{i=1}^Q$: instead of taking the mean of all targets like CART does, it chooses a constant value c that minimizes the mean Hinge Error (2) between this constant and the target intervals:

$$\underset{c}{\operatorname{argmin}} \frac{1}{Q} \sum_{i=1}^{Q} l\big(c, [\log \lambda_l^i, \log \lambda_u^i]\big), \quad c \in \mathbf{R}$$

A heuristic algorithm is used to speed up the estimation of the mean value for each leaf by considering a finite set of candidates, rather than solving for the optimal c:

$$\underset{c}{\operatorname{argmin}} \frac{1}{Q} \sum_{i=1}^{Q} l\big(c, [\log \lambda_l^i, \log \lambda_u^i]\big),$$
$$\text{where } c \in \{\log \lambda_l^i, \log \lambda_u^i, \frac{\log \lambda_l^i + \log \lambda_u^i}{2} | \log \lambda_l^i > -\infty, \log \lambda_u^i < \infty\}_{i=1}^Q$$

The training procedure for determining the optimal tree architecture follows the same operations as in CART (binary optimization) to minimize the Hinge Error (2) over the train dataset having S instances $\{\mathbf{x}^i, [\log \lambda_l^i, \log \lambda_u^i]\}_{i=1}^S$:

$$\sum_{i=1}^{S} l\Big(\log \hat{\lambda}^i, [\log \lambda_l^i, \log \lambda_u^i]\Big) = \sum_{i=1}^{S} l\Big(\mathcal{T}(\mathbf{x}^i), [\log \lambda_l^i, \log \lambda_u^i]\Big)$$

The optimal tree architecture (maximum depth, minimum split), loss type which is p (either 1 or 2) in Hinge Error (2), and margin which is ϵ in Hinge Error (2), is selected through cross-validation.

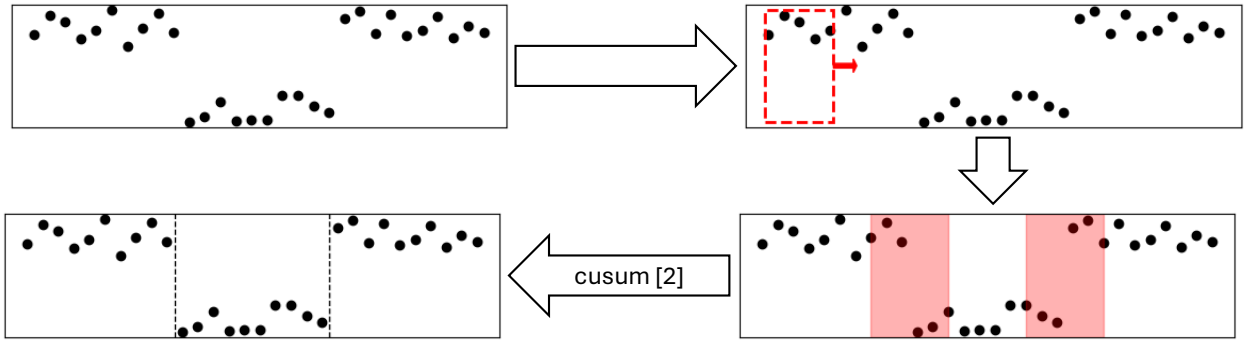


Figure 5: Changepoint detection using Deep Learning

2 Literature Review: Optimal Partitioning

2.1 Changpoint Detection Algorithms

Changepoint detection serves as a vital tool in numerous real-life applications by pinpointing significant transitions or sudden changes in data patterns, ranging from detecting market trends in finance [Lattanzi and Leonelli, 2021], monitoring disease outbreaks in healthcare [Muggeo and Adelfio, 2010], enhancing network security [Tartakovsky et al., 2013], monitoring environmental changes [Reeves et al., 2007] and more.

Changepoint detection methods can be divided into two categories: Deep Learning (DL) and Dynamic Programming (DP). DL-based methods, which inspired from a statistical method Cumulative Sum (CUSUM) [Hinkley, 1971, James et al., 1987], including recent works [Li et al., 2024, Ermshaus et al., 2023] employ multilayer perceptrons (MLP) or k-nearest neighbors (KNN) as classification sliding windows to identify changepoints. Take a look at the Figure 5 for a clear visualization of the operation. However, these methods rely on local neighbors rather than the entire sequence and are overly complex for univariate data. In contrast, DP is well-suited for this context.

To determine the optimal partition of a sequence given a fixed number of segments (or number of changepoints), several efficient algorithms can be employed to achieve precise results [Auger and Lawrence, 1989, Bai and Perron, 2003, Killick et al., 2012]. However, in many instances, the number of segments is not predetermined and must instead be inferred from the data. To address this

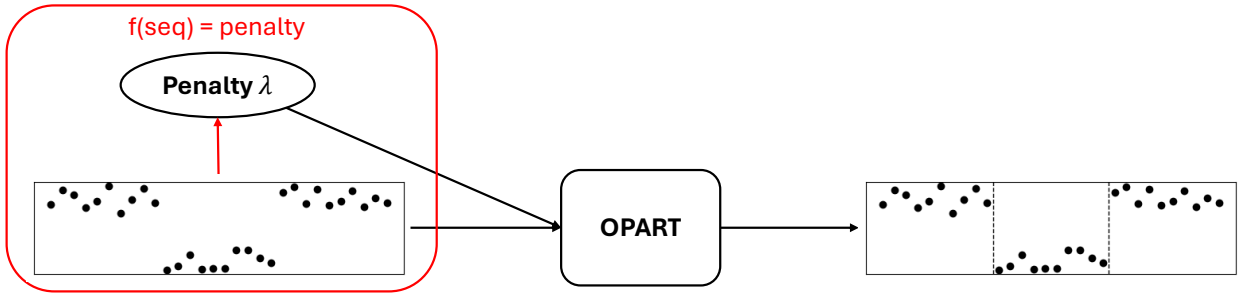


Figure 6: The challenge of using OPART is predicting the value for penalty

challenge, several dynamic programming algorithms have been successfully applied to changepoint detection, including Optimal Partitioning (OPART) [Jackson et al., 2005] with its variations such as Pruned Exact Linear Time (PELT) [Killick et al., 2012] or Functional Pruning Optimal Partitioning (FPOP) [Maidstone et al., 2017]. PELT uses a theorem to prune certain candidate changepoints that cannot be part of the optimal partition. FPOP prunes at least as many candidate changepoints as PELT and can be more efficient, particularly for large datasets. Despite these differences, OPART, PELT, and FPOP all produce the same partition. The Labeled Optimal Partitioning (LOPART) [Hocking and Srivastava, 2023] extends OPART's capabilities by leveraging predefined changepoint labels, indicating the expected number of changepoints within specific location ranges. When no such labels are defined, LOPART behaves identically to OPART. Without prior knowledge of the number of changepoints in the partition and aiming to limit their occurrence, these algorithms employ a penalty parameter to discourage the presence of changepoints, denoted as λ, in the following optimization problem

OPART problem Find partition vector $\mathbf{m} \in \mathbb{R}^N$ that minimizes the following cost function for a given sequence $\mathbf{d} \in \mathbb{R}^N$ and penalty parameter $\lambda \geq 0$:

$$\mathbf{m} = \underset{\mathbf{m}}{\operatorname{argmin}}\, C_\lambda(\mathbf{d}, \mathbf{m}) \tag{5}$$

$$\text{where} \quad C_\lambda(\mathbf{d}, \mathbf{m}) = \sum_{i=1}^{N} l(d_i, m_i) + \lambda I[m_i \neq m_{i+1}] \tag{6}$$

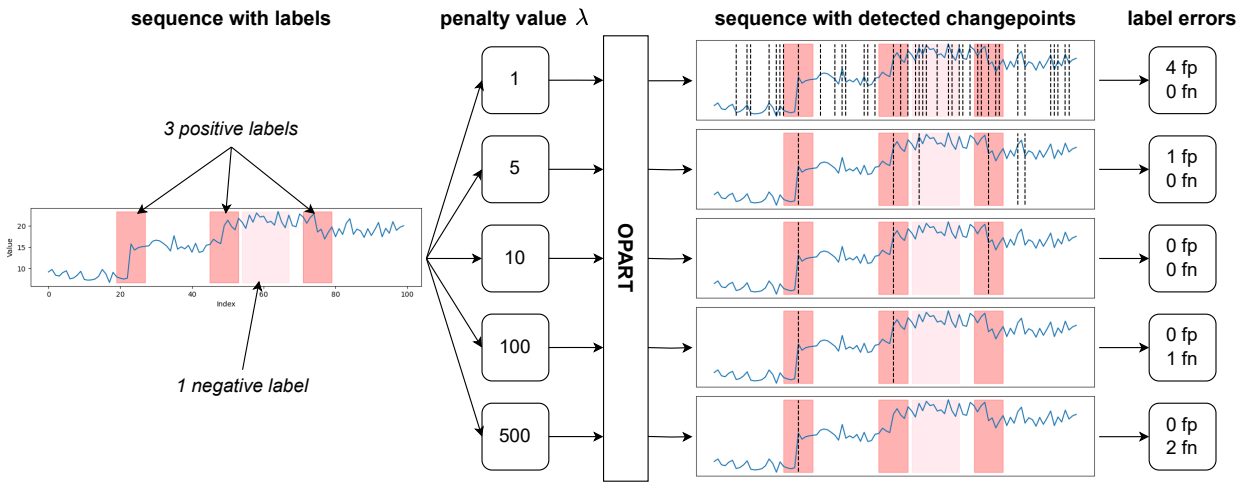


Figure 7: Different penalty result in varied sets of changepoints

Note: Example: how λ value influences changepoints count. The labeled sequence has 3 positive labels (each with one changepoint) and 1 negative label (with no changepoints). Label errors are classified as false positives (fp), occurring when multiple changepoints are detected in a positive label or any changepoint is detected in a negative label, and false negatives (fn), occurring when no changepoint is detected in a positive label. In this example, $\lambda = 10$ is optimal.

where $l(m_i, d_i)$ usually represents the squared error between the sequence value d_i and the mean segment m_i, i.e., $l(d_i, m_i) = (m_i - d_i)^2$. $I[m_i \neq m_{i+1}]$ is an indicator function that equals 1 if there is a changepoint (when $m_i \neq m_{i+1}$) and 0 otherwise. Thus, $k = \sum_{i=1}^{N} I[m_i \neq m_{i+1}]$ represents the total number of changepoints, hence $k + 1$ is the number of segments. For instance, for a sequence $\mathbf{d} = [d_1, d_2, d_3, d_4, d_5, d_6] \in \mathbb{R}^6$, if we have a fixed penalty λ^*, the optimal partition vector of problem 5 is $\mathbf{m} = [m_1, m_1, m_1, m_1, m_2, m_2]$, this indicates that the partition $\mathbf{m}$ consists of 1 changepoint and 2 segments: $[d_1, d_2, d_3, d_4]$ and $[d_5, d_6]$.

Role of the penalty parameter λ Within the OPART problem 5, the penalty parameter λ holds significant importance in partitioning. Moreover, a higher λ imposes a more substantial penalty for changepoints' presence, consequently yielding smaller sets of changepoints. When $\lambda = 0$, the optimal partition $\mathbf{m}$ in problem 5 matches the given sequence $\mathbf{d}$ exactly ($\mathbf{m} \equiv \mathbf{d}$), while as λ approaches $+\infty$, all values in $\mathbf{m}$ is the mean of $\mathbf{d}$. A too high value of λ is undesirable because it results in fewer detected changepoints, potentially missing significant ones. Conversely, a too low

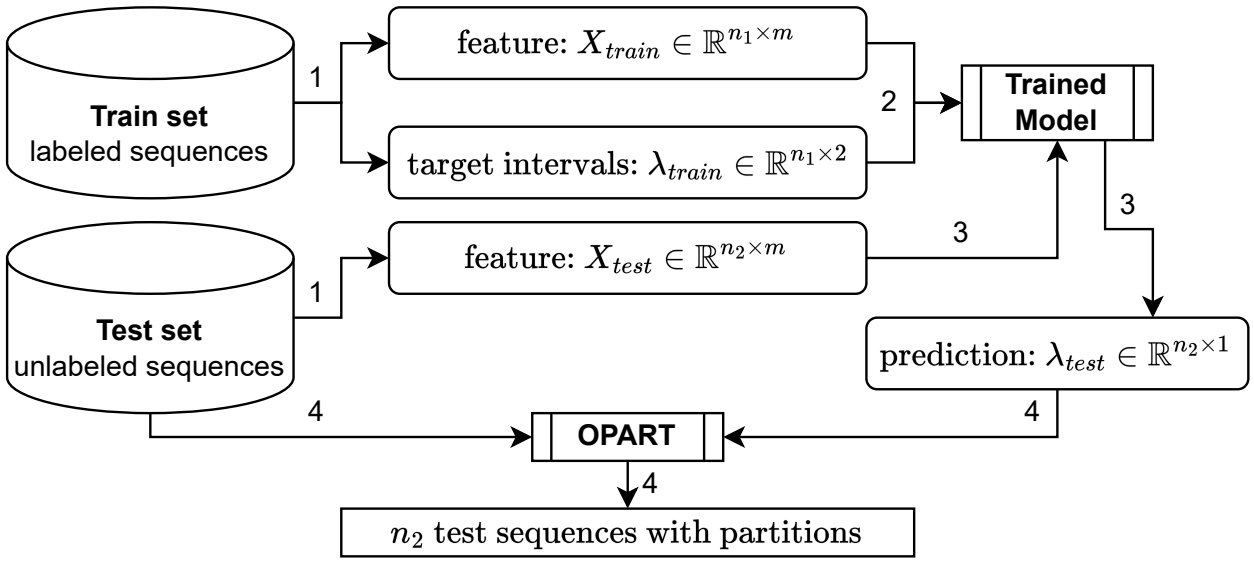


Figure 8: The process involves a labeled sequences train set and a unlabeled sequences test set

Note: a train set with n_1 labeled sequences and a test set with n_2 unlabeled sequences. Step 1: Extract m features from both sets and get a target interval from each labeled sequence in the train set. Step 2: Train the interval regression model using the features and target intervals from the train set. Step 3: Use the trained model to predict λ for each test sequence. Step 4: For each unlabeled sequence in the test set, apply OP using the predicted λ obtained from the trained model.

value of λ results in an excessive number of detected changepoints, more than necessary. While the existing changepoint detection algorithms has demonstrated effectiveness, the fixed nature of λ prompts inquiry into methods for dynamically adapting this critical parameter.

2.2 Relationship between OPART and Interval Regression

OPART algorithms take into the sequence $\mathbf{d}$ along with λ as input to generate the set of detected changepoints. Each labeled sequence $\mathbf{d}^j$ has M_j distinct labels $\{(s_i, e_i, c_i)\}_{i=1}^{M_j}$, where c_i denotes the expected number of changepoints between start location s_i and end location e_i. There are two types of label errors: in each label, if the number of detected changepoints greater than c_i, it's a *false positive*; if smaller, it's a *false negative*.

Finding the optimal partition for a given labeled sequence require considering a range of values for λ (rather than just one specific value) that minimizes the number of label errors. Refer to [Rigaill et al., 2013] for a method to determine the optimal range of λ for each labeled sequence.

Consequently, the value of λ to be predicted should fall within an optimal interval, see Figure 1. This type of problem, where the objective is to find values falling within an interval, is referred to as the *interval regression*.

3 Literature Gaps and Proposed Plans

3.1 Inteval Regression

Limitation Let me briefly summarize the previous works and highlight their limitations:

- **Original AFT model and its variants**: Designed for survival analysis, it handles only uncensored and right-censored data, as it does not account for left-censored data.
- **AFT model in XGBoost**: Capable of handling all types of censored data by transforming target intervals to ensure non-negative values. It introduces a loss function for all types of censored outcomes. However, being tree-based, the model lacks smooth decision boundaries.
- **Linear model in supervised machine learning**: Similar to the original AFT model, it has a simple linear architecture that struggles with datasets exhibiting non-linear patterns.
- **MMIT**: A non-linear model that captures complex relationships, but like any tree-based models, it produces non-smooth decision boundaries, leading to a lack of smoothness in predictions.

Proposed Plan My plan involves utilizing more sophisticated and smoother models, such as Generalized Additive Models (GAMs) [Hastie, 2017], random forest, or like multi-layer perceptrons (MLP).

- One option would be an MLP with just one hidden layer and very few neurons, using Rectified Linear Unit activation functions like ReLU or Leaky ReLU (not ELU or SELU or anything that creates non-linear regions). For instance, an MLP with one hidden layer and two neurons, using Leaky ReLU as the activation function, can be about twice as complex as a linear model, as it can create up to four linear regions (see Proposition 3 in [Montufar et al., 2014]). This makes it a more general approach than traditional linear models without being overly complex.

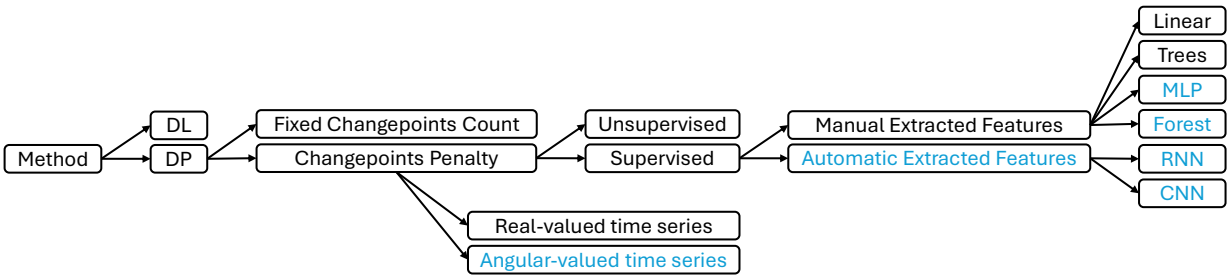


Figure 9: Comprehensive Diagram: Review and Contributions

- Alternatively, for a non-linear model that remains relatively simple, I could opt for piecewise linear models, like splines [Friedman, 1991] with linear terms in each region.

3.2 Changepoint Detection

Limitation The OPART algorithm employs a fixed penalty value, λ, which may limit its adaptability to different data structures.

Proposed Plan

- Explore heuristic approaches that divide the data into smaller segments, followed by applying optimal partitioning to each subsequence.
- Consider applying k-means clustering to detect potential segments.
- Applying complex models for penalty prediction.

4 Paper: Optimal Partitioning Penalty Learning using MLP

Changepoint detection (or other words, partitioning and segmentation) is essential in many real-world applications, identifying significant data shifts in finance [Lattanzi and Leonelli, 2021], healthcare [Muggeo and Adelfio, 2010], network security [Tartakovsky et al., 2013], environmental monitoring [Reeves et al., 2007], and more.

Detecting changepoints involves selecting the detection method and tuning hyperparameters to detect the changepoint locations.

Changepoint detection methods review Many real-life time series data are univariate. Several deep learning (DL)-based changepoint detection methods exist [Londschien et al., 2023, Lee et al., 2023, Li et al., 2024, Ermshaus et al., 2023], these approaches generalize classical methods [Hinkley, 1971, James et al., 1987] by applying sliding classification windows built using either multilayer perceptron (MLP) or k-nearest neighbors (KNN). DL-based methods are fairly easy to understand and implement, making them accessible to a wide audience. But these methods require tuning numerous hyperparameters, making them overly complex for univariate data. In contrast, dynamic programming (DP) algorithms provides an ideal solution in this context. *This study selects DP to enhance its changepoint detection performance.*

Regularization for Changepoints Count in DP Algorithms DP algorithms detects changepoints using only one hyperparameter (contrast to numerous hyperparameters in DL-based methods) to control changepoints count, either (a) a fixed changepoints count or (b) a penalty to penalize changepoint presence. Approach (a), used in [Auger and Lawrence, 1989, Bai and Perron, 2003, Killick et al., 2012], is effective but relies on often unknown changepoints count. Approach (b), with the most popular methods being Optimal Partitioning (OP) [Jackson et al., 2005] and its variants, including Pruned Exact Linear Time (PELT) [Killick et al., 2012], Functional Pruning Optimal Partitioning (FPOP) [Maidstone et al., 2017], and Labeled Optimal Partitioning (LOP) [Hocking and Srivastava, 2023]. PELT and FPOP improves efficiency by pruning technique whilst producing the

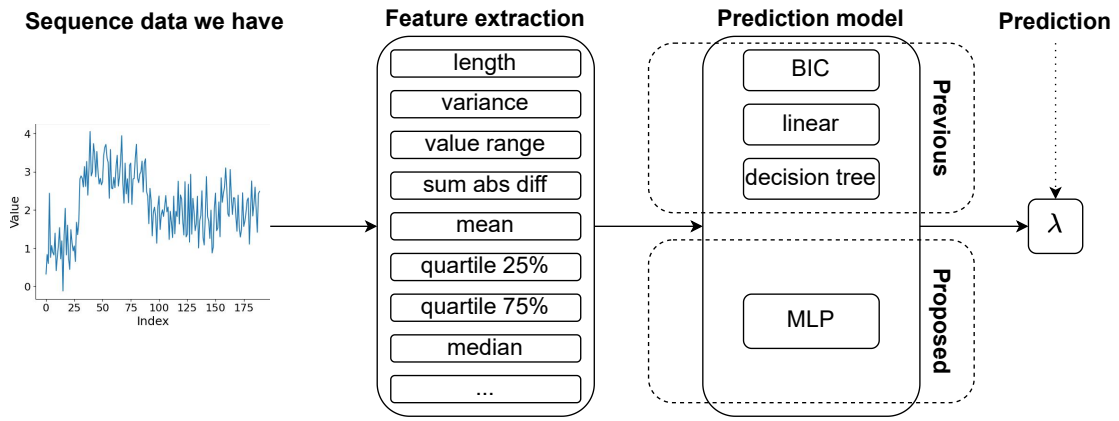


Figure 10: Comparative Model Diagram - Previous Methods versus Our Proposal

same solution as OP. LOP extends OP by incorporating predefined changepoint labels and defaults to OP when labels are absent. *This study selects approach (b) to enhance changepoint detection performance as it offers greater flexibility.*

The role of the OP penalty A higher penalty imposes stricter penalties, reducing the changepoints count (see Figure 7). Given the suitability of OP family algorithms for changepoint detection in univariate sequences and the role of the OP penalty, *this study focuses on developing a model to predict the optimal OP penalty*.

Related Work review Various methods exist for predicting the OP penalty, including unsupervised [Schwarz, 1978, Akaike, 1974, Lavielle, 2005] and supervised ones [Truong et al., 2017, Rigaill et al., 2013, Drouin et al., 2017, Barnwal et al., 2022].

Gap in the Literature and Contributions Existing models are constrained by simplistic architectures, such as linear or tree-based, which motivated us to propose a novel approach. This study proposes MLP with activation function ReLU for OP penalty prediction. This is a generalized version of a linear model and offers continuous nonlinear predictions compared to tree-based models. Experiments conducted on three large genome datasets demonstrate that the proposed model outperforms previous work in accuracy and F1 score. For simplicity, we refer to the OP penalty as λ

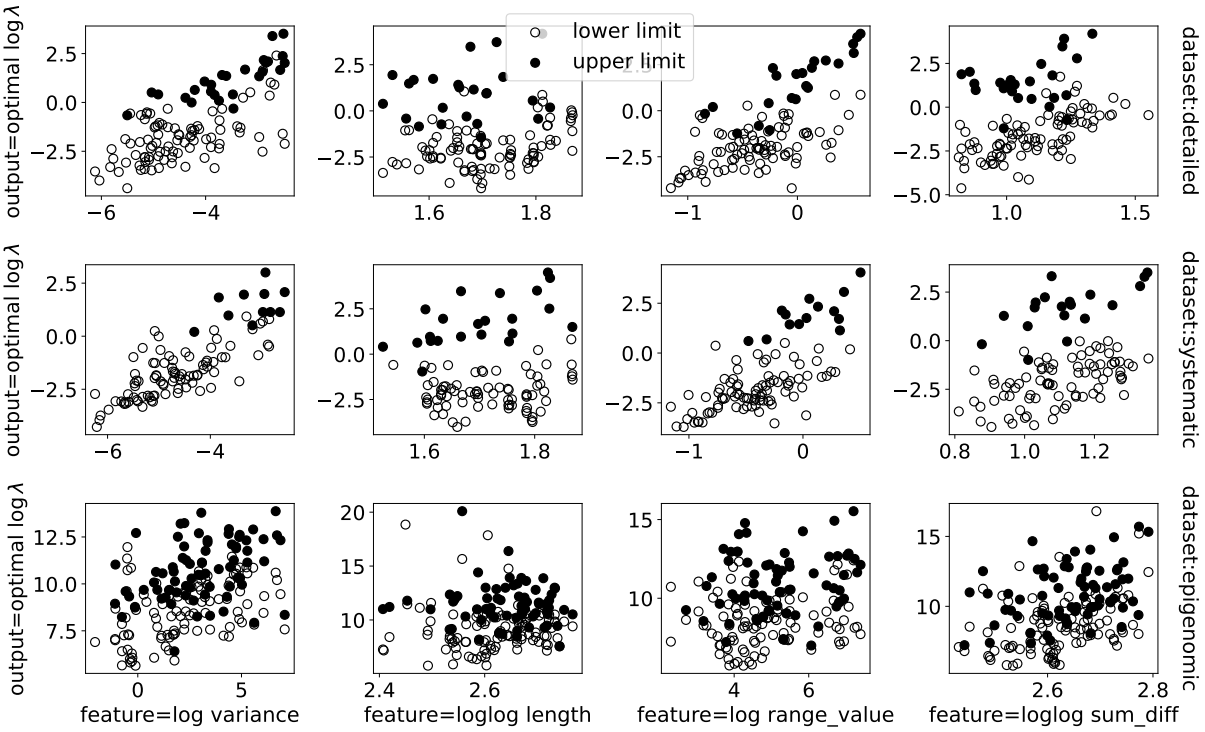


Figure 11: Graphs showing the relationship between four features and target intervals

Graphs illustrating the relationship between four features and target intervals. Variance and length are selected because previous studies have incorporated them into the model. Upon examination, we observe that the range value and sum of absolute differences exhibit a discernible increasing monotonic linear relationship with the target interval, indicating their potential as good features.

hereafter.

4.1 Previous Work

Noteworthy Adaptive Linear Penalty INference (ALPIN) [Truong et al., 2017] is a method for a special case of this supervision, where every point in the sequence is a label, equivalent to label regions having length 1. Since this study uses a more relaxed supervision with varying label region lengths, ALPIN is not applicable. Accelerated Failure Time (AFT) models, a class of models for censored outcomes, can be considered. While changepoint detection datasets may include various interval types, traditional AFT models [Cai et al., 2009, Huang et al., 2006, Breiman, 2001, Pölsterl et al., 2016] are limited to uncensored and right-censored intervals, making them inapplicable.

Model	Features	Model Type	Regularization	Citation
BIC.1	1	unsupervised	None	Schwarz [1978]
linear.1 linear.2 linear.4	1 2 4	linear	None	Hocking and Srivastava [2023] Rigaill et al. [2013] Lavielle [2005]
linear.all	118		L1	Rigaill et al. [2013]
mmit.1 mmit.2 mmit.4 mmit.all	1 2 4 118	tree	max depth min split sample loss type loss margin	Drouin et al. [2017]
aft_xgboost.1 aft_xgboost.2 aft_xgboost.4 aft_xgboost.all	1 2 4 118	ensemble tree	distribution scale learning rate max depth min child weight reg alpha reg lambda	Barnwal et al. [2022]
mlp.1 mlp.2 mlp.4 mlp.all	1 2 4 118	MLP	hidden layers count neurons per layer early stopping	proposed

Table 4: List of employed penalty prediction models

Unsupervised The Bayesian Information Criterion (BIC) [Schwarz, 1978] sets $\lambda = \log N$, where N is the sequence length, while Akaike's Information Criterion (aic) [Akaike, 1974] selects $\lambda = 2p$, with p is a feature (e.g, variance). Lavielle [2005] does not specify a particular form but computes λ based on sequence length and variance.

Supervised The AFT in the XGBoost model [Barnwal et al., 2022], despite being an AFT model, can handle all non-negative target intervals, making it suitable for this study.

Since λ is non-negative, to eliminate this constraint in models output, the following models predict $\log \lambda$ instead of λ: linear [Rigaill et al., 2013] and Maximum Margin Interval Trees (MMIT) [Drouin et al., 2017]. Both are trained by using a specific loss function, the Hinge Error. Using the ReLU function $\big(f(x) = \max(0, x)\big)$, Hinge Error between prediction $\hat{y}$ and target interval $[y_l, y_u]$ is expressed as:

$$l(\hat{y}, [y_l, y_u]) = \big(\text{ReLU}(y_l - \hat{y} + \epsilon)\big)^p + \big(\text{ReLU}(\hat{y} - y_u + \epsilon)\big)^p \tag{7}$$

Here, $\epsilon \geq 0$ is the margin length (with [Rigaill et al., 2013] using $\epsilon = 1$ and [Drouin et al., 2017] treating ϵ as a hyperparameter), and p (1 or 2) defines the loss type. It is clear that if the prediction $\hat{y}$ falls within the interval $[y_l + \epsilon, y_u - \epsilon]$, the Hinge Error value is 0.

4.2 Contribution

Novelty: MLP for λ prediction A MLP with fully connected architecture is used. All hidden layers have the same number of neurons. ReLU activation function is used in the input and hidden layer(s). This architecture makes the proposed model a more general form of a linear model and produces continuous nonlinear predictions compared to tree-based models like MMIT and AFT in XGBoost.

Additional Contribution: Feature Selection A meaningful feature distinguishes sequences from one another in the context of changepoint detection. Features such as the mean, quartile values (min, Q1, Q2, Q3, max) and their transformations are not selected because these features are noise in the changepoint context. For example, $[0, 0, 0, 1, 1, 1]$ and $[1, 1, 1, 2, 2, 2]$ are the same sequences. Sequence length and variance alone cannot fully represent a sequence, since some sequences may have the same length and variance but differ significantly in this context, so we introduce two additional sequence-related features: *value range* and *sum of absolute differences*.

- **Value range:** This feature (r in Table 8) is defined as the difference between the maximum and minimum values in the sequence.

- **Sum of absolute differences:** This feature (s in Table 8) is calculated as the sum of the absolute differences between two consecutive sequence points, denoted as $\sum_{i=1}^{N-1} |d_{i+1} - d_i|$, where $\mathbf{d} = [d_1, d_2, \ldots, d_N]$ is the sequence. This feature differentiates sequences with

Dataset	Sequences	Pos. Labels	Neg. Labels	Source
Detailed	3730	964	3396	GitHub
Systematic	3418	573	2846	GitHub
Epigenomic	4913	2185	8195	UCI Repo

Table 5: Datasets used in the study

the same length, variance, or value range, such as [0,0,0,1,1,1] and [0,1,0,1,0,1], which are distinct in the context of changepoint detection.

Intuitively, as the value range or the sum of absolute differences increases, the sequence tends to show more fluctuations, indicating a need to increase λ. See Figure 11 to see why these two features emerge as strong candidates.

I employed cross-validation to determine the appropriate MLP architecture, which was time-consuming, for details, see Figure 14. My approach achieved improved accuracy rates in change-point detection when applied to benchmark datasets (Figure 12). I have uploaded my research to arXiv: link. And it's published in Statistics and Computing journal: Spring Nature link.

4.3 Experiments and Results

Features Four sets of features are considered to evaluate their quality:

- **1 feature**: Sequence length, as used in Hocking and Srivastava [2023] and Schwarz [1978].
- **2 features**: Sequence length and variance, following Rigaill et al. [2013].
- **4 features**: Sequence length, variance, value range, sum of absolute differences.
- **All features**: According to Rigaill et al. [2013] and Drouin et al. [2017], up to 365 features are generated, with only those without of missing data, infinite values, or zero variance retained.

Models Configuration All models employed 5-fold cross-validation to select the optimal hyper-parameters.

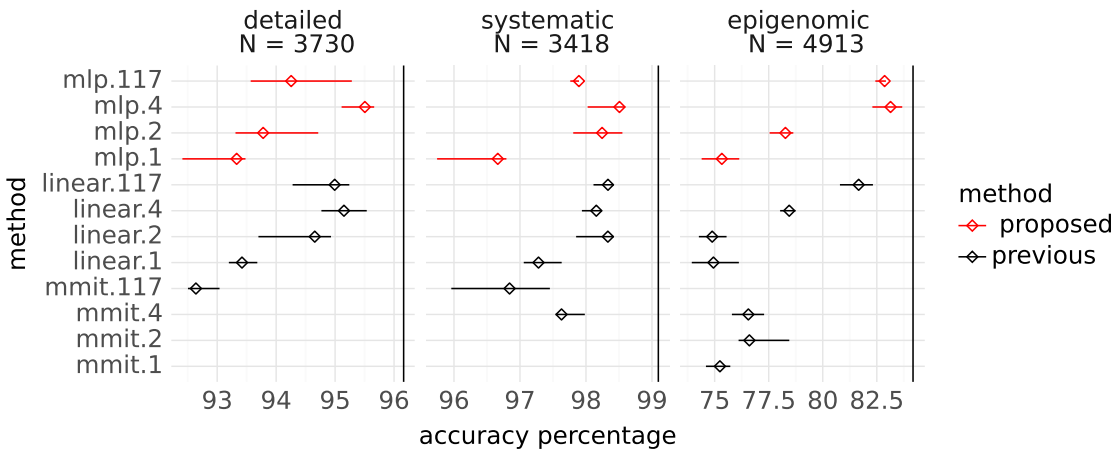


Figure 12: Median test accuracy and interquartile across 6 folds for each method

- **Linear**: L1 regularization is applied only to the large feature vector, with cross-validation used to determine the optimal L1 parameter value, starting at 0.001 and increasing by 1.2 until no features remain.

- **MMIT**: The hyperparameters considered include a list for `max_depth` values (0, 1, 5, 10, 20, Inf), `margin` (0, 1, 2) (ϵ from equation 2), `loss_type` (`hinge`, `square` – equivalent to $p = 1$ or $p = 2$ in Equation 2) and `min_sample` (0, 1, 2, 4, 8, 16, 20).

- **AFT in XGBoost**: The hyperparameters considered include a list of `learning_rate` values (0.001, 0.01, 0.1, 1.0), `max_depth` (2, 3, 4, 5, 6, 7, 8, 9 10), `min_child_weight` (0.001, 0.1, 1.0, 10.0, 100.0), `reg_alpha` (0.001, 0.01, 0.1, 1.0, 10.0, 100.0), `reg_lambda` (0.001, 0.01, 0.1, 1.0, 10.0, 100.0), `aft_loss_distribution_scale` (0.5, 0.8, 1.1, 1.4, 1.7, 2.0). This setup is identical to that described in [Barnwal et al., 2022].

- **MLP**: The hyperparameters considered include a list of `n_hidden_layer` (1, 2, 3, 4) and `hidden_layer_size` (2, 4, 8, 16, 32, 64, 128, 256, 512). We used the early stopping technique using a `patience` parameter, stopping training if the loss does not decrease after the specified patience iterations; in our study, we set the maximum iterations to 12,000 and the patience to 20. The model is trained using the Adam optimizer (with default PyTorch

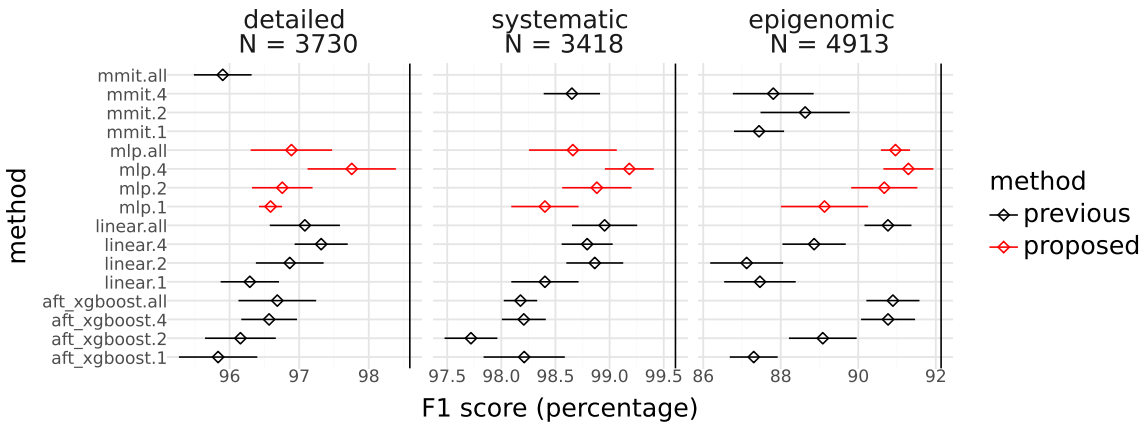


Figure 13: Median test F1 score and interquartile across 6 folds for each method.

hyperparameters) to minimize the Squared Hinge Loss (as defined in Function 2 with $p = 2$ and $\epsilon = 1$).

Results This study uses 3 large datasets, as detailed in Table 4.2. Each dataset is split into 6 similar size folds. In each iteration, 1 fold is designated as the test set, and the remaining 5 folds form the train set, with the process yield 6 test accuracy rates (Figure 12) and F1 scores (Figure 13) for each method.

4.4 Discussion and Conclusion

Performance Analysis and Overall Comparison Figures 12 and 13 demonstrate that the proposed model, when trained with the selected four-feature configuration, consistently achieves comparable or slightly superior performance relative to previously published linear and tree-based approaches across all evaluated datasets. Although the observed improvements in mean accuracy and F1 score are modest, they are stable across cross-validation folds, indicating that the gains are systematic rather than incidental.

In particular, increasing the number of features from one to two and then to four generally leads to improved predictive performance for all model families considered. This trend suggests that the selected features provide complementary information that enhances the model's ability to correctly

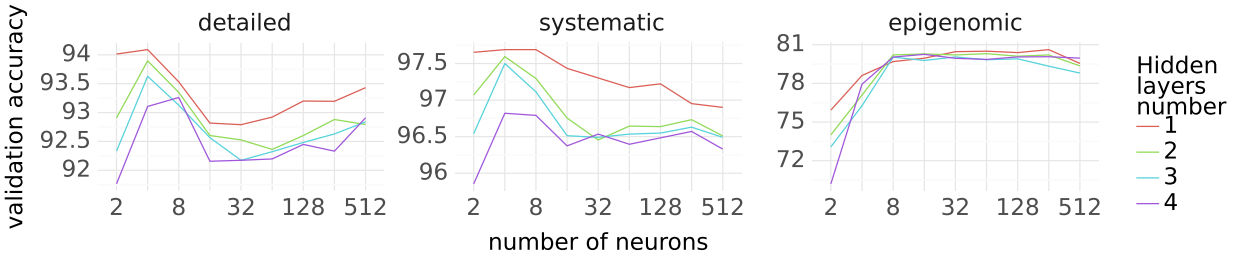


Figure 14: Average validation accuracies for different MLP configurations

identify changepoint intervals. However, the inclusion of all available features does not necessarily yield further improvement. In several cases, performance stagnates or slightly deteriorates when all features are used, indicating the presence of redundant or weakly informative variables. This highlights the importance of careful feature selection, especially for flexible models such as MLPs.

Comparison Between Model Families A notable observation is that tree-based models do not consistently outperform linear models when trained on the same feature sets. This phenomenon aligns with findings reported in [Drouin et al., 2017] for the Systematic dataset and in [Barnwal et al., 2022] for subsets of the Epigenomic dataset. The primary reason appears to be the predominantly linear relationship between the extracted features and the target intervals, as illustrated in Figure 11. When the underlying structure of the data is largely linear, more complex nonlinear learners such as tree ensembles do not offer a significant advantage.

The proposed model, formulated as a piecewise linear MLP, is conceptually positioned between purely linear models and highly nonlinear deep architectures. As expected, for identical feature configurations, it generally outperforms the standard linear model. The improvement can be attributed to its ability to capture mild nonlinearities while preserving much of the interpretability and stability of linear approaches. Importantly, strong performance is achieved with a relatively simple architecture—typically a single hidden layer containing 4 or 8 neurons—demonstrating that excessive architectural complexity is unnecessary for these datasets.

Consistency and Stability of the Proposed Model While the proposed model achieves strong mean performance, Figures 12 and 13 also indicate that MLPs tend to exhibit wider confidence intervals compared to linear and tree-based models. This variability likely stems from the higher sensitivity of neural networks to initialization, optimization dynamics, and hyperparameter configurations. Despite this increased variance, the overall performance remains competitive and robust across folds. The results suggest that moderate architectural simplicity contributes to improved stability without sacrificing predictive power.

Factors Affecting Performance

- *Feature Design.* The experiments confirm that feature quality plays a more critical role than model complexity. The relationship between sequence length and the target interval of λ, as shown in Figure 11, is not clearly separable, which may explain why models trained solely on this feature underperform relative to those incorporating additional statistical descriptors such as variance or range. The fact that nonlinear models do not benefit from the inclusion of all features further underscores that feature relevance, rather than feature quantity, determines performance gains.

- *MLP Configuration.* Determining an effective MLP configuration required systematic experimentation. We evaluated 36 distinct configurations for each train–test pair by varying the number of hidden layers (1–4) and the number of neurons per layer (2 to 512, in powers of two). This grid search was computationally demanding. Nevertheless, results indicate that shallow architectures with limited width are sufficient. Given the largely linear structure of the problem, deeper or wider networks did not provide consistent improvements, reinforcing the principle that model capacity should match data complexity.

Limitations Several limitations should be acknowledged. First, feature extraction was performed manually, and only a limited subset of possible statistical descriptors was considered. The space

of potential features derivable from sequences is effectively unbounded; measures such as skewness, kurtosis, entropy, or counts of repeated values were not included. Consequently, potentially informative predictors may have been overlooked.

Second, unlike L1-regularized linear models or tree-based algorithms, MLPs lack an inherent feature selection mechanism. This limitation reduces flexibility when handling larger feature spaces and increases the risk of incorporating redundant variables. As a result, careful preprocessing remains essential for neural network models in this setting.

Third, identifying optimal MLP hyperparameters is time-consuming and computationally intensive. Although an extensive search was conducted, configurations with heterogeneous neuron counts across layers were not explored, leaving open the possibility that alternative architectures could yield marginal improvements.

Conclusion and Future Directions Overall, the proposed piecewise linear MLP model achieves comparable or slightly improved accuracy and F1 scores in changepoint detection relative to traditional linear and tree-based models. The results emphasize that thoughtful feature selection is more influential than architectural complexity. While the model benefits from moderate nonlinear flexibility, its performance does not justify highly complex network designs for the datasets studied.

Future research may focus on automating feature learning rather than relying on manual extraction. In particular, sequence-oriented neural architectures such as Recurrent Neural Networks (RNNs) [Rumelhart et al., 1986], Gated Recurrent Units (GRUs) [Chung et al., 2014], and Long Short-Term Memory (LSTM) networks [Hochreiter, 1997] could be investigated for directly learning informative representations from raw sequences. Such approaches may alleviate the need for handcrafted features and potentially enhance generalization to other sequence domains.

Reproducible Research Material To promote transparency and facilitate replication, all source code and experimental materials are publicly available at: github. Providing open access to the implementation ensures that results can be independently verified and extended by future researchers.

5 Paper: Optimal Partitioning Penalty Learning using RNN

Changepoint detection (other words, partitioning or segmentation) is important for identifying abrupt changes in data in various domains, including finance [Lattanzi and Leonelli, 2021], healthcare [Muggeo and Adelfio, 2010], network security [Tartakovsky et al., 2013], and environmental science [Reeves et al., 2007]. This importance has driven the development of numerous methods over the past decades.

Changepoint Detection Methods. Changepoint detection methods can be divided into two categories: Deep Learning (DL) and Dynamic Programming (DP).

DL-based methods, which inspired from a statistical method Cumulative Sum (CUSUM) [Hinkley, 1971, James et al., 1987], including recent works [Li et al., 2024, Ermshaus et al., 2023] employ multilayer perceptrons (MLP) or k-nearest neighbors (KNN) as classification sliding windows to identify changepoints. However, these methods rely on local neighbors rather than the entire sequence and are overly complex for univariate data.

In contrast, DP is well-suited for this context. These algorithms rely on a single hyperparameter to limit the changepoints number, which can be either: (a) a fixed changepoints number or (b) a penalty parameter to penalize the changepoint presence. Fixed changepoints number allow DP algorithms [Auger and Lawrence, 1989, Bai and Perron, 2003, Killick et al., 2012, Rigaill, 2015] to locate them, but the changepoints number is rarely predetermined. More commonly used, penalty-based DP algorithms, such as Optimal Partitioning (OPART) [Jackson et al., 2005] and its variants—Pruned Exact Linear Time (PELT) [Killick et al., 2012], Functional Pruning Optimal Partitioning (FPOP) [Maidstone et al., 2017], and Labeled Optimal Partitioning (LOPART) [Hocking and Srivastava, 2023]—are widely regarded as the most effective. PELT and FPOP prune candidate changepoints efficiently, producing identical partitions as OPART, while LOPART extends OPART by incorporating predefined labels (the expected number of changepoints within specific location ranges) or defaulting to OPART when labels are absent. The following is the OPART optimization

problem:

OPART Problem. Given a sequence $\mathbf{d} \in \mathbb{R}^N$ and penalty parameter $\lambda \geq 0$, find the vector $\mathbf{m} \in \mathbb{R}^N$ that minimizes the following cost function:

$$C(\mathbf{d}, \mathbf{m}) + \lambda M$$

Here, $C(\mathbf{d}, \mathbf{m})$ represents the error between the sequence $\mathbf{d}$ and the vector $\mathbf{m}$ and M is the number of changes in $\mathbf{m}$. The formula of $C(\mathbf{d}, \mathbf{m})$ and M depends on the type of change. If the type of change is mean, then $C(\mathbf{d}, \mathbf{m}) = \sum_{i=1}^{N}(m_i - d_i)^2$ and $M = \sum_{i=1}^{N-1} I[m_i \neq m_{i+1}]$, where $I[m_i \neq m_{i+1}]$ equals 1 if a changepoint occurs ($m_i \neq m_{i+1}$) and 0 otherwise.

In OPART algorithms, the penalty plays a key role. A higher penalty imposes a stronger penalty on changepoint presence, leading to fewer changepoints, see Figure 7. This study aims to predict that penalty to improve the changepoint detection accuracy. From here on, that conceptual penalty will be referred to as λ.

Existing λ prediction models. Various models exist for predicting λ, including unsupervised [Schwarz, 1978, Akaike, 1974, Lavielle, 2005] and supervised such as linear [Rigaill et al., 2013] and tree(s) [Drouin et al., 2017, Barnwal et al., 2022]. But all of these models rely on a set of manually extracted sequence features, yet the number of possible features is infinite, this process can result in the inclusion of many irrelevant features, and the potential omission of useful ones.

Contribution. This study proposes a novel method utilizing a recurrent network to automatically extract relevant features from sequences, which are then used to predict λ. Experiments on benchmark datasets show that the proposed method achieves higher partitioning accuracy than existing ones in most cases.

Reproducibility. To ensure transparency, the code used for this study is accessible at: `https://github.com/lamtung16/ML_Changepoint_Detection_epigenomic_rnn`.

5.1 Previous Work

This subsection focuses on the process of training supervised λ prediction models. Before going into the details, some noteworthy models are highlighted, which, although not applicable to this study, are still relevant. Additionally, some unsupervised models, which have served as inspiration for supervised models, are mentioned.

Noteworthy. Adaptive Linear Penalty INference (ALPIN) [Truong et al., 2017] is a method for this supervision special case, where every point is a label, equivalent to label regions having length 1. However, since this study uses a more relaxed supervision with varying label region lengths, ALPIN is not applicable. The Accelerated Failure Time (AFT) model [Wei, 1992] and its variants [Cai et al., 2009, Huang et al., 2006, Quinlan, 1986, Pölsterl et al., 2016], which are designed for censored outcomes, can be considered. Supervised changepoint detection setups involve interval-censored targets, whereas traditional AFT models are restricted to uncensored or right-censored intervals, making them not directly applicable.

Unsupervised. These models predicts λ based on some sequence features. Let $\hat{\lambda}$ denote a prediction of λ. Let N denote the sequence length, and σ its standard deviation, with examples such as Baysian Information Criterion (BIC) [Schwarz, 1978] predicts $\hat{\lambda} = \sigma^2 \log N$. Akaike's Information Criterion (AIC) [Akaike, 1974] predicts $\hat{\lambda} = 2p$, where p represents a sequence feature (e.g., standard deviation, variance). Lavielle [2005] does not provide a closed-form formula but considers $\hat{\lambda}$ to be dependent on both N and σ.

Supervised. The rest of this section will detail the training process of three supervised machine learning models: Maximum Margin Interval Regression [Rigaill et al., 2013], a linear model; Max-

imal Margin Interval Tree (MMIT) [Drouin et al., 2017]; and AFT in XGBoost [Barnwal et al., 2022], which utilizes a tree-based framework (although AFT in XGBoost is an AFT model, it is applicable due to its ability to handle all types of non-negative intervals). These models divided the training process into 2 sequential independent steps:

- Step 1: Extract features vector $\mathbf{x}$ from the sequence.
- Step 2: Train a model $g(\cdot)$ takes into $\mathbf{x}$ to predict $\hat{\lambda} = g(\mathbf{x})$

5.1.1 Step 1: Feature Extraction

The manual feature extraction process, detailed in Rigaill et al. [2013], involves the following steps:

- Starting from the sequence $\mathbf{d} = [d_1, d_2, \ldots, d_N]$ with the mean value $\bar{d} = \frac{1}{N}\sum_{i=1}^{N} d_i$, 2 additional sequences (residuals and differences) are generated:

$$\mathbf{d}_{\text{res}} = [d_1 - \bar{d}, \ldots, d_N - \bar{d}] \in \mathbb{R}^N$$
$$\mathbf{d}_{\text{diff}} = [d_2 - d_1, \ldots, d_N - d_{N-1}] \in \mathbb{R}^{N-1}$$

- Apply 3 transformations (identity, absolute, square) to each of 3 vectors, resulting in 9 transformed vectors.
- Compute 8 statistics (sum, mean, standard deviation, and quantiles at 0%, 25%, 50%, 75%, 100%) from each transformed vector, yielding a vector of length 72.
- Include the sequence length as an additional feature, forming a vector of length 73.
- Apply 5 transformations (identity, square root, log, log-log, square) to each feature, resulting in up to 365 features.

The values caused by invalid operations (e.g., log of a negative number), is ignored.

5.1.2 Step 2: Model Training

After obtaining the feature vector $\mathbf{x}$ (up to 365 features), each instance $\{\mathbf{d}, [\lambda_l, \lambda_u]\}$ becomes $\{\mathbf{x}, [\lambda_l, \lambda_u]\}$, a selected model g is trained to predict $\hat{\lambda} = g(\mathbf{x})$.

AFT in XGBoost. The model is formulated as:

$$\hat{\lambda} = \mathbf{T}(\mathbf{x}) + \varepsilon$$

Here, $\mathbf{T}(\mathbf{x})$ denotes the ensemble of trees for feature vector $\mathbf{x}$, and ε is a random error. The model is trained by minimizing the negative log-likelihood over all train instances:

$$l_{AFT}\Big(\hat{\lambda}, [\lambda_l, \lambda_u]\Big) = -\log\Big[F_Z\big(\frac{\log\lambda_u - \log\hat{\lambda}}{\sigma}\big) - F_Z\big(\frac{\log\lambda_l - \log\hat{\lambda}}{\sigma}\big)\Big] \tag{8}$$

where $\sigma > 0$ is the scale parameter. The cumulative distribution function $F_Z(z)$ depends on the chosen distribution of ε:

- normal: $F_Z(z) = \frac{1}{2}\big(1 + \text{erf}(\frac{z}{\sqrt{2}})\big)$
- logistic: $F_Z(z) = \frac{e^z}{1+e^z}$
- extreme: $F_Z(z) = 1 - e^{-\exp z}$

The model is trained using the gradient boosting algorithm [Friedman, 2001] to minimize the loss value (8) over the training dataset with S instances $\{\mathbf{x}^i, [\lambda_l^i, \lambda_u^i]\}_{i=1}^S$ given the distribution of ε and the scale parameter σ:

$$\sum_{i=1}^{S} l_{AFT}\Big(\hat{\lambda}^i, [\lambda_l^i, \lambda_u^i]\Big) = -\sum_{i=1}^{S}\log\Big[F_Z\big(\frac{\log\lambda_u^i - \log\mathbf{T}(\mathbf{x}^i)}{\sigma}\big) - F_Z\big(\frac{\log\lambda_l^i - \log\mathbf{T}(\mathbf{x}^i)}{\sigma}\big)\Big]$$

Linear. This model, known as Maximum Margin Interval Regression, transforms the problem to predicting $\log\hat{\lambda}$ instead of $\hat{\lambda}$, as $\log\hat{\lambda}$ can be any real value, unlike the non-negative $\hat{\lambda}$. The model

expresses the output as a linear combination of inputs:

$$\log \hat{\lambda} = \mathbf{x} \cdot \beta + \beta_0 \tag{9}$$

where β is the parameter vector, $\mathbf{x} \cdot \beta$ is the dot product of $\mathbf{x}$ and β, and β_0 is the bias term. Its simplified version, used in Hocking and Srivastava [2023], predicts $\log \hat{\lambda} = \beta_1 \times \log \log N + \beta_0$, where N is the sequence length.

This model is trained by minimizing a loss function, the Hinge Error, defined by Rigaill et al. [2013], it achieves 0 loss when the prediction falls within the target interval. The error value between prediction $\log \hat{\lambda}$ and target interval $[\log \lambda_l, \log \lambda_u]$ is:

$$l\Big(\log \hat{\lambda}, [\log \lambda_l, \log \lambda_u]\Big) = \begin{cases} (\log \lambda_l - \log \hat{\lambda} + \epsilon)^2 & \text{if } \log \hat{\lambda} < \log \lambda_l + \epsilon, \\ 0 & \text{if } \log \lambda_l + \epsilon \leq \log \hat{\lambda} \leq \log \lambda_u - \epsilon, \\ (\log \hat{\lambda} - \log \lambda_u + \epsilon)^2 & \text{if } \log \hat{\lambda} > \log \lambda_u - \epsilon. \end{cases}$$

where $\epsilon \geq 0$ is the margin length ($\epsilon = 0$ by default). The Hinge Error becomes Square Error when $\lambda_l = \lambda_u$ and $\epsilon = 0$ (or $\epsilon = \frac{\log \lambda_l + \log \lambda_u}{2}$), or in other word, target interval becomes target point. Drouin et al. [2017] proposed a similar loss function:

$$l\Big(\log \hat{\lambda}, [\log \lambda_l, \log \lambda_u]\Big) = \begin{cases} |\log \lambda_l - \log \hat{\lambda} + \epsilon| & \text{if } \log \hat{\lambda} < \log \lambda_l + \epsilon, \\ 0 & \text{if } \log \lambda_l + \epsilon \leq \log \hat{\lambda} \leq \log \lambda_u - \epsilon, \\ |\log \hat{\lambda} - \log \lambda_u + \epsilon| & \text{if } \log \hat{\lambda} > \log \lambda_u - \epsilon. \end{cases}$$

Using the ReLU function, the general loss function can be written in this short form:

$$l(\log \hat{\lambda}, [\log \lambda_l, \log \lambda_u]) = \Big(\text{ReLU}(\log \lambda_l - \log \hat{\lambda} + \epsilon)\Big)^p + \Big(\text{ReLU}(\log \hat{\lambda} - \log \lambda_u + \epsilon)\Big)^p, \tag{10}$$

where $\epsilon \geq 0$ is the margin length and $p \in \{1, 2\}$ is the loss type.

To estimate the parameters β and β_0 in (11), the Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) [Beck and Teboulle, 2009] with respect to β and β_0 is employed to minimize the Hinge Error (10) over the train dataset having S instances $\{\mathbf{x}^i, [\log \lambda_l^i, \log \lambda_u^i]\}_{i=1}^S$:

$$\sum_{i=1}^{S} l\Big(\log \hat{\lambda}^i, [\log \lambda_l^i, \log \lambda_u^i]\Big) = \sum_{i=1}^{S} l\Big(\mathbf{x^i} \cdot \beta + \beta_0, [\log \lambda_l^i, \log \lambda_u^i]\Big)$$

Tree. MMIT introduced by Drouin et al. [2017] is applicable for λ prediction as an interval regression model. The MMIT model is:

$$\log \hat{\lambda} = \mathcal{T}(\mathbf{x})$$

The tree $\mathcal{T}$ architecture follows the same structure as CART (Classification And Regression Tree) [Breiman et al., 1984]. The only difference lies in the regression value for each leaf which is a set of Q instances $\{\mathbf{x}^i, [\log \lambda_l^i, \log \lambda_u^i]\}_{i=1}^Q$. Instead of taking the mean of all targets like CART does, MMIT chooses a constant value c that minimizes the mean Hinge Error (10) between this constant and the target intervals:

$$\underset{c}{\operatorname{argmin}} \frac{1}{Q} \sum_{i=1}^{Q} l\big(c, [\log \lambda_l^i, \log \lambda_u^i]\big), \quad c \in \mathbf{R}$$

A heuristic algorithm is used to speed up the estimation of the mean value for each leaf by considering a finite set of candidates, rather than solving for the optimal c:

$$\underset{c}{\operatorname{argmin}} \frac{1}{Q} \sum_{i=1}^{Q} l\big(c, [\log \lambda_l^i, \log \lambda_u^i]\big),$$
$$\text{where } c \in \{\log \lambda_l^i, \log \lambda_u^i, \frac{\log \lambda_l^i + \log \lambda_u^i}{2} | \log \lambda_l^i > -\infty, \log \lambda_u^i < \infty\}_{i=1}^Q$$

The training procedure for determining the optimal tree architecture follows the same operations as

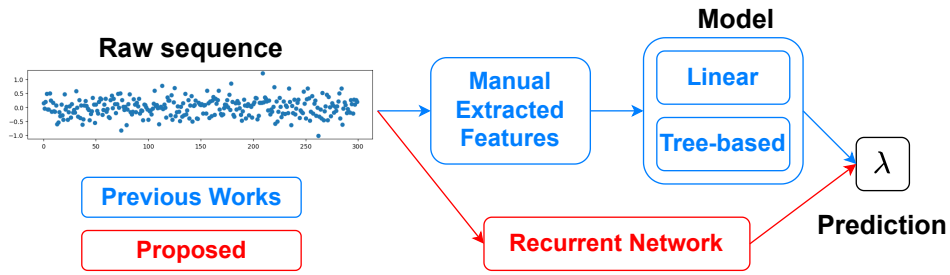


Figure 15: Diagram of the proposed methods using RNN

in CART (binary optimization) to minimize the Hinge Error (10) over the train dataset having S instances $\{\mathbf{x}^i, [\log \lambda_l^i, \log \lambda_u^i]\}_{i=1}^S$:

$$\sum_{i=1}^{S} l\Big(\log \hat{\lambda}^i, [\log \lambda_l^i, \log \lambda_u^i]\Big) = \sum_{i=1}^{S} l\Big(\mathcal{T}(\mathbf{x}^i), [\log \lambda_l^i, \log \lambda_u^i]\Big)$$

5.2 Contribution

Feature Extraction via Recurrent Networks. In a Recurrent Network with input size $k \geq 1$ and hidden size $m \geq 1$, processing a sequence $\mathbf{d} = [d_1, d_2, \ldots, d_N] \in \mathbb{R}^N$, the hidden state at time t, called h_t, is updated as:

$$h_t = f(d_{[t:t+k-1]}; h_{t-1}) \in \mathbb{R}^m$$

where $d_{[t:t+k-1]} = [d_t, d_{t+1}, ..., d_{t+k-1}] \in \mathbb{R}^k$. Time stamp t ranges from k to N. The initial hidden state h_{k-1} can be assigned manually. The final hidden state $h_N \in \mathbb{R}^m$ captures information from the entire sequence. Thus, we use h_N as the m extracted features from the raw sequence.

Predicting λ from extracted features. The prediction is formed by the linear combination of m extracted features:

$$\log \lambda = h_N \cdot \beta + \beta_0$$

where $\beta \in \mathbb{R}^m$ and $\beta_0 \in \mathbb{R}$ are the parameters to be learned.

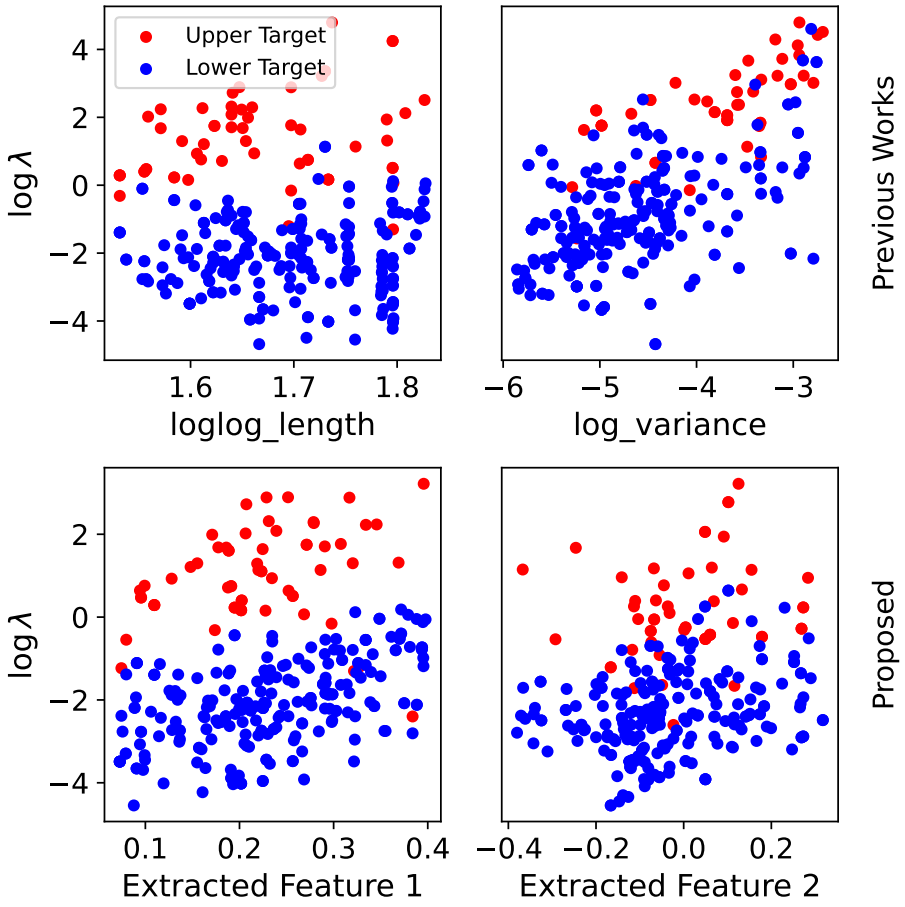


Figure 16: Example: features vs. targets from the dataset detailed

The upper plots: the length or variance vs. targets, as these features are used on λ prediction unsupervised methods. The lower plots: automatically extracted features vs targets, using a GRU model with two extracted features. Since we consider the predicted λ has a linear relationship with the features, a good feature should exhibit a fairly linear pattern in the plot.

Automatic extracted features vs. Manual extracted features. The novel method generates useful features without the need for a large number, unlike previous models. Figure 16 provides an example by comparing the ability to predict λ using length (or variance) - as these manual extracted features are considered key for λ prediction in unsupervised methods - versus automatic extracted features from recurrent networks. It demonstrates that the automatic extracted features probably provide a more accurate prediction of λ than either length or variance.

Recurrent core: $h_t = f(d_{[t-k+1:t]}; h_{t-1}) \in \mathbb{R}^m$				
sequence feature	k	m	h_{k-1}	h_t
length	$k \in \mathbf{N}^+$	1	$k-1$	$h_{t-1}+1$
mean	1	1	0	$h_{t-1} + \frac{d_t - h_{t-1}}{t}$
min	1	1	∞	$\min(d_t, h_{t-1})$
max	1	1	$-\infty$	$\max(d_t, h_{t-1})$
sum	1	1	0	$h_{t-1} + d_t$
cumulative absolute	1	1	0	$h_{t-1} + \lvert d_t \rvert$
cumulative squares	1	1	0	$h_{t-1} + d_t^2$
difference sum	2	1	0	$\lvert d_t - d_{t-1} \rvert + h_{t-1}$
value range	1	2	∞ $-\infty$	$\min(d_t, h_{t-1}^{(1)})$ $\max(d_t, h_{t-1}^{(2)})$

Table 6: Formula of recurrent core for each feature approximation.

Feature Approximation Ability of Recurrent Networks. Recurrent networks have the ability to approximate statistical sequence features. For example, if the core network is defined as $h_t = h_{t-1} + 1$ when $h_0 = 0$ and the input size $k = 1$, then $h_N = N$, which represents the sequence length, a feature used in BIC. The core component of these models is a deep neural network. For instance, the Vanilla Recurrent Network [Rumelhart et al., 1986] core is a MLP, which, according to the universal approximation theorem [Hornik et al., 1989], can approximate any continuous function, regardless of its complexity, provided there are enough neurons in the hidden layer. This is why, although these models may not directly produce exact statistical sequence features, their architecture enables them to approximate these features with a high degree of accuracy.

Recurrent networks to be implemented. This study explores three recurrent networks: Vanilla Recurrent Network (RNN), Long Short-Term Memory (LSTM) [Hochreiter, 1997], Gated Recurrent Unit (GRU) [Chung et al., 2014]. RNNs struggle with long-term dependencies due to the vanishing gradient problem. LSTMs address this by introducing memory cells and gating mechanisms that help retain information over longer sequences. GRUs, a simplified version of LSTMs,

Dataset	Seq. length	Size	Source
systematic	66 - 5937	3418	GitHub
detailed	25 - 5937	3730	
cancer	39 - 43628	826	
ATAC_JV_adipose	105716 - 11.50M	465	UCI Repo
CTCF_TDH_ENCODE	166431 - 11.09M	182	
H3K27ac-H3K4me3_TDHAM_BP	275 - 4.33M	2008	
H3K27ac_TDH_some	41053 - 6.74M	95	
H3K27me3_RL_cancer	5730 - 1.89M	171	
H3K27me3_TDH_some	113523 - 3.00M	43	
H3K36me3_AM_immune	184092 - 7.14M	420	
H3K36me3_TDH_ENCODE	68209 - 2.74M	78	
H3K36me3_TDH_immune	434882 - 4.35M	37	
H3K36me3_TDH_other	49362 - 7.15M	40	
H3K4me1_TDH_BP	148275 - 9.73M	144	
H3K4me3_PGP_immune	24460 - 6.62M	297	
H3K4me3_TDH_ENCODE	18198 - 7.21M	75	
H3K4me3_TDH_immune	87557 - 7.81M	378	
H3K4me3_TDH_other	80674 - 3.72M	90	
H3K4me3_XJ_immune	52498 - 5.50M	270	
H3K9me3_TDH_BP	369654 - 10.29M	120	

Table 7: List of datasets with its number of sequences (Size)

combine the forget and input gates into a single update gate, offering a more efficient alternative while still capturing long-term dependencies.

5.3 Experiments and Results

In this section, the implementation of models is detailed. Labeled sequences are first processed, with features manually extracted to replicate previous work, along with the corresponding target intervals. Baseline models and the proposed models are then implemented. For each test sequence, OPART is applied with the predicted λ to detect changepoints. Finally, accuracy rates are calculated based on the predicted changepoints and the test set labels.

Raw Sequence Datasets. This study uses 20 datasets: two DNA copy number profiles from neuroblastoma tumors [Rigaill et al., 2013], known for detailed and systematic dataset; one dataset cancer from [Hocking and Srivastava, 2023]; and 17 ChIP-seq datasets from [Hocking et al., 2016], see Table 4.2 for all benchmark datasets, each sequence has a different length (number of datapoints), so the range of sequence lengths within each dataset is reported in the Table 4.2. For each sequence in each dataset, biologist experts use their visual interpretation of important signal and noise patterns to label regions as having changes in mean value or not.

Train/Test Setup. Each sequence is assigned a unique identifier (sequenceID), and datasets are split into folds based on these IDs. Fewer folds are used for smaller datasets to balance train and test data sizes, ensuring a sufficiently large test set for reliable evaluation. In each iteration, one fold serves as the test set, and the others form the train set.

Evaluation Metrics. The metric used is the accuracy rate on the test set:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where

- TP = True Positives (The number of detected changepoints matches to the positive labels)
- TN = True Negatives (No detected changepoints in the negative labels)
- FP = False Positives (The number of detected changepoints exceeds that of the labels.)
- FN = False Negatives (The number of detected changepoints is less than that of the labels.)

5.3.1 Model Implementation

Previous models were implemented as baselines, along with additional models and the proposed models for comparison. Details of all models are provided in Table 8.

Table 8: List of employed models

Model	Regularization	Language	Package	Paper
linear	L1	R	[Hocking, 2024]	Rigaill et al. [2013]
MMIT	tree architecture	Python/Cpp	[Drouin et al., 2017]	Drouin et al. [2017]
AFT_XGboost	tree architecture	Python	[Chen and Guestrin, 2016]	Barnwal et al. [2022]
constant	none	Python/Cpp	[Drouin et al., 2017]	Baseline
MLP	layer number layer sizes	Python		Additional
RNN LSTM GRU	layer number state sizes	Python		Proposed

5.3.2 Baseline Models

Linear. This model uses L1 regularization. The regularization parameter L1 starts at 0.001, geometrically increasing by a factor of 1.2 until no features remain, with cross-validation determining the optimal L1 regularization value.

MMIT. The optimal hyperparameters are chosen through cross-validation on the training set, including:

- `max_depth`: 0, 1, 5, 10, 20, and ∞
- `min_sample`: 0, 1, 2, 4, 8, 16, and 20.
- `margin` (ϵ in Hinge Error 10): 0, 1, and 2
- `loss_type`: `hinge` and `square` (equivalent to $p = 1$ and $p = 2$ in Hinge Error 10);

AFT in XGBoost. Following the same setup described in Barnwal et al. [2022], the optimal hyperparameters are chosen via cross-validation on the training set, including:

- `learning_rate`: 0.001, 0.01, 0.1, and 1.0
- `max_depth`: 2, 3, 4, 5, 6, 7, 8, 9, and 10

- `min_child_weight`: 0.001, 0.1, 1.0, 10.0, and 100.0
- `reg_alpha`: 0.001, 0.01, 0.1, 1.0, 10.0, and 100.0
- `reg_lambda`: 0.001, 0.01, 0.1, 1.0, 10.0, and 100.0
- `aft_loss_distribution_scale`: 0.5, 0.8, 1.1, 1.4, 1.7, and 2.0

Constant (Featureless). This model predicts a single value $\hat{\lambda} = \exp \log \hat{\lambda}$ from the train set using only target intervals. The objective is to minimize the total Hinge Error (10) over S instances with target intervals $[\log \lambda_l^i, \log \lambda_u^i]_{i=1}^S$:

$$\log \hat{\lambda} = \underset{c}{\operatorname{argmin}} \sum_{i=1}^{S} l(c, [\log \lambda_l^i, \log \lambda_u^i]).$$

5.3.3 Additional Model

MLP. Using the same feature vector and loss function as the linear model, the MLP generalizes the linear model by incorporating ReLU activation in the hidden layers. Hyperparameters include the number of hidden layers (1, 2, or 3) and the size of each hidden layer (1, 2, 4, 8, 16, 32, 64, 128, 256, or 512). The best hyperparameters are chosen via cross-validation in the train set. Since the linear model uses L1 regularization and MMIT (or AFT in XGBoost) has a tree-based architecture, they can automatically select features. However, MLP does not fully mitigate irrelevant features on its own (parameters associated with each feature are likely non-zero). Feature selection for MLP in this interval regression setting could be explored in future work.

5.3.4 Proposed Models

Architecture. There are three considered models: RNN, LSTM, and GRU. The hyperparameters include: the number of hidden layers (1 or 2) and the hidden state size (number of extracted features) (2, 4, 8 or 16).

Preprocessing. Based on the datasets sequence lengths, we handle them as follows:

- For the 3 datasets—detailed, systematic, cancer—the sequence lengths range from 39 to 43628. These sequences are directly input into the λ prediction models without further modification.

- For the 17 ChIP-seq datasets, the sequence lengths can be over 11 million. Training on these raw sequences can be computationally expensive and time-consuming. So pooling operations are performed, where a single representative value is chosen for each non-overlapping sliding window. The window size options are 100, 1000, or 2000, and the representative value is either the window mean or the median. After shortening the sequences, a $\log(z+1)$ transformation is applied to all value z in all shortened sequences to mitigate skewed distributions. Following this, all sequence values are normalized (mean 0, variance 1) for each dataset.

5.3.5 Results

The comprehensive accuracy results of all 8 methods for each of 20 datasets are provided in the Appendix, as the table is too large to be included in the main paper. Figure 17 presents the accuracy rates for three datasets using the original, unpooled sequences, while Figure 18 shows the results for various ChIP-seq datasets (pooled sequences). Each dataset is evaluated using seven different methods, producing multiple accuracy values depending on the number of folds used for that dataset.

5.4 Discussion and Conclusion

Summary of Contributions. Technically, these proposed models still perform two steps: (1) Feature extraction and (2) Predicting λ based on the extracted features. However, instead of performing these steps separately, it combines them into a single operation, enhancing its robustness compared to previous models. Automatic extracting features helps generate highly useful ones without requiring many features, as shown in Figure 16. This study focuses exclusively on genomic datasets,

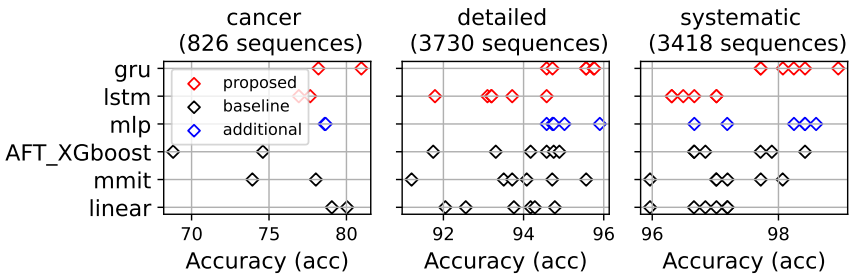


Figure 17: The accuracies for each fold are presented for datasets without any pooling

where prior knowledge of relevant features enables effective manual feature extraction. However, the proposed models stand out because it does not require expert knowledge to perform. Its automatic feature extraction mechanism makes it versatile and applicable to any type of sequence dataset.

Comparison to Previous Models. Recurrent networks, in general, outperform previous models across datasets, see Figure 17. For small datasets (with fewer than 100 instances), the accuracy of the proposed models is comparable (neither better nor worse) to that of existing models. On the other hand, when dealing with larger datasets, recurrent networks noticeably outperform previous models and show a slight edge over MLPs.

Comparison between Recurrent Networks. When comparing RNN, LSTM, and GRU, as expected, RNN performs the worst due to its difficulty with long-term memory. In most cases, GRU outperforms the others.

Discussion on the Number of Automatically Extracted Features. The number of automatic extracted features needs to be sufficiently large. It is observed that when 8 or 16 features are extracted, the validation loss value is slightly smaller compared to when only 2 or 4 features are extracted.

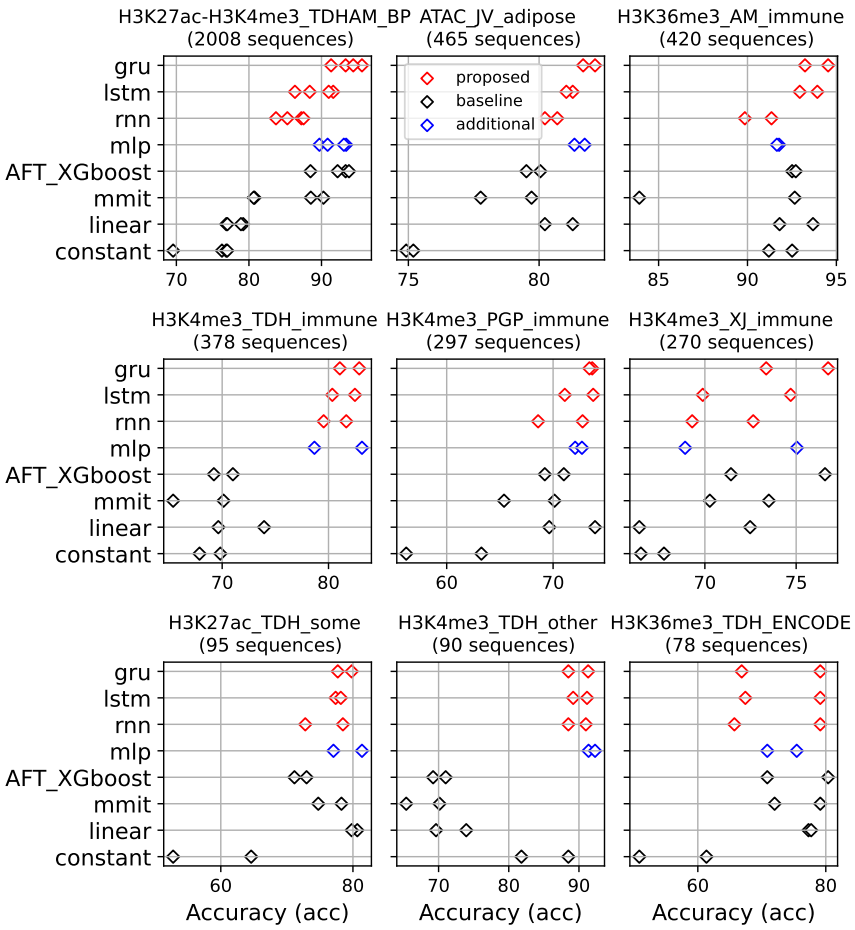


Figure 18: The fold accuracies on some ChIP-seq datasets

Impact of pooling preprocessing. Shortening the sequence for training is effective. As the window size increases, more information is lost. However, when the window size is small, such as 100, the performance of the recurrent models is worse compared to larger window sizes like 1000 or 2000. This could be because the pooled sequence with a window size of 100 is still too long for the recurrent models to handle effectively. The pooling process is beneficial in two ways: it can improve the accuracy of recurrent networks and also reduce training time. We employed two pooling functions, mean and median, to represent a single value for each non-overlapping window. It is evident that using the mean value as the representation consistently yields smaller loss.

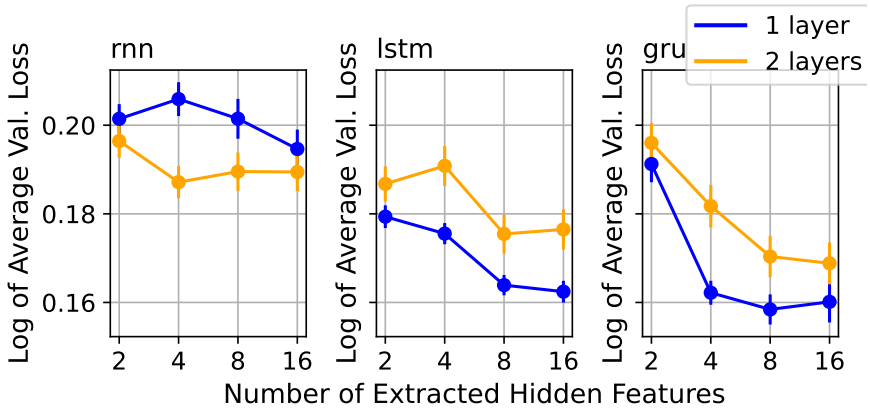


Figure 19: Log of the loss validation across different numbers of extracted hidden layers

Study Limitations. Although the Recurrent Network architecture is straightforward, as its core essentially a deep neural network, training these models can be time-consuming due to the inherently sequential nature of the process. For some train/test pairs, even after applying pooling to reduce the sequence length, training a model with a maximum of 2 layers, each containing no more than 16 neurons, and a maximum of 1000 iterations, took over approximately 10 days, despite utilizing the powerful NVIDIA A100 GPU. These models also require significant memory to store the hidden states at each time step. To avoid out-of-memory (OOM) errors, it is crucial to allocate sufficient memory, especially for long sequences.

Practical Considerations in Real-World Applications. There is a trade-off between training time and accuracy. The proposed models have higher accuracy in most of the 20 datasets, but they are significantly more time-consuming to train, especially for longer sequences. In contrast, manually extracted features models do not face this issue. Therefore, in real-world scenarios where datasets contain only a few sequences but each sequence is long, and training time is a priority, manually extracted features models may be the better choice for two reasons: first, their training time is significantly faster; and second, these models' accuracy is comparable to the proposed models. However, if the user's priority is accuracy and the hardware requirements are met, the proposed

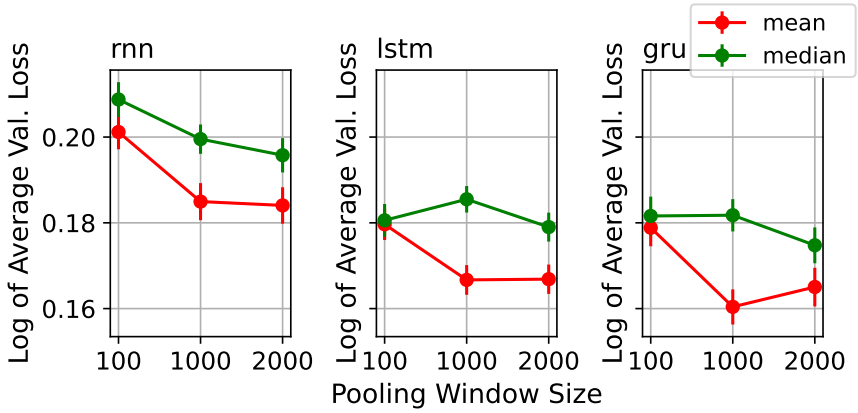


Figure 20: Log of the loss validation across different window size

models are a better choice. Although the training process is time-consuming, it only needs to be done once; afterwards, the trained models can be applied to unseen sequences, as their feed-forward process is sufficiently fast.

Future Directions. For these particular datasets, it is assumed that λ exhibits an increasingly monotonic behavior with respect to both sequence length and variance, which are employed in unsupervised methods. One approach is to leverage monotonic network architectures, such as min-max networks [Sill, 1997], or more advanced architectures like Lattice Networks [You et al., 2017], with modifications to the loss function for each linear region.

We can consider CNNs [LeCun et al., 1998] because they are capable of handling sequences of varying lengths and generating a single output by utilizing convolution layers to extract features across the sequence. Global pooling layers (e.g., Global Max or Average Pooling) aggregate these features into a fixed-size representation. This vector is then passed through a MLP to produce the final output.

To build upon the initial concept of utilizing recurrent networks in this study, various types of models could be explored in the future:

- Using a more complex model: This study predicts λ by using a linear combination of the

extracted features. Alternatively, we could consider using a more complex model such as MLP that processes the extracted features instead of relying on a linear combination.

- Attention mechanisms [Bahdanau et al., 2014] dynamically focus on relevant input parts, assigning importance weights to subsequences. Unlike sequential RNNs, they effectively capture long-range dependencies and handle varying input lengths, making them ideal for this task.

- Bidirectional Recurrent Networks [Schuster and Paliwal, 1997]: The models implemented in this study are unidirectional, leading to challenges with long-term memory, since the later parts of the sequence tend to be more important than the earlier ones. Using a bidirectional recurrent network architecture can aim to address this limitation, enabling the model to better capture information from the sequence.

Conducting even a preliminary experiment would require substantial computing resources, since bidirectional networks and attention mechanisms are more computationally expensive than the proposed one. For this reason, we chose not to include such experiments as part of this work.

Table 9: Accuracy means and standard deviations (by method)

Dataset	constant	linear	mmit	AFT_XGB	mlp	rnn	lstm	gru
cancer	63.37 ± 14.93	79.55 ± 0.70	75.97 ± 2.90	71.70 ± 4.10	78.62 ± 0.04	75.42 ± 2.12	77.29 ± 0.52	79.58 ± 1.96
detailed	67.45 ± 6.42	93.60 ± 1.07	93.80 ± 1.47	93.91 ± 1.20	94.95 ± 0.49	86.26 ± 4.64	93.26 ± 0.91	95.32 ± 0.53
systematic	56.87 ± 9.87	96.81 ± 0.46	97.16 ± 0.72	97.37 ± 0.75	97.92 ± 0.80	72.87 ± 29.01	96.64 ± 0.32	98.19 ± 0.47
ATAC_JV_adipose	75.05 ± 0.19	80.75 ± 0.75	78.74 ± 1.38	79.78 ± 0.38	81.55 ± 0.28	80.46 ± 0.33	81.16 ± 0.17	81.91 ± 0.32
CTCF_TDH_ENCODE	75.47 ± 1.71	84.15 ± 0.40	87.65 ± 0.54	93.32 ± 0.84	93.96 ± 0.38	86.58 ± 0.23	89.97 ± 2.50	91.52 ± 7.80
H3K27ac-H3K4me3_TDHAM_BP	74.93 ± 3.60	78.00 ± 1.22	85.06 ± 5.06	91.95 ± 2.41	91.76 ± 1.77	85.94 ± 1.78	89.33 ± 2.44	93.65 ± 1.80
H3K27ac_TDH_some	58.69 ± 8.35	80.23 ± 0.60	76.51 ± 2.48	72.06 ± 1.30	79.21 ± 3.05	75.63 ± 4.03	77.77 ± 0.56	78.76 ± 1.49
H3K27me3_RL_cancer	54.86 ± 1.35	64.52 ± 5.54	65.90 ± 2.32	65.23 ± 0.78	67.58 ± 3.57	64.43 ± 3.32	65.49 ± 3.24	64.22 ± 4.08
H3K27me3_TDH_some	62.19 ± 2.89	85.48 ± 1.90	77.19 ± 6.68	85.40 ± 3.35	77.83 ± 5.26	85.43 ± 2.36	87.21 ± 2.26	86.66 ± 2.01
H3K36me3_AM_immune	91.85 ± 0.93	92.75 ± 1.33	88.29 ± 6.18	92.61 ± 0.15	91.72 ± 0.09	90.60 ± 1.06	93.44 ± 0.70	93.89 ± 0.92
H3K36me3_TDH_ENCODE	56.10 ± 7.42	77.51 ± 0.29	75.57 ± 5.05	75.61 ± 6.73	73.16 ± 3.25	72.43 ± 9.49	73.28 ± 8.28	73.00 ± 8.69
H3K36me3_TDH_immune	96.39 ± 1.96	87.78 ± 11.00	87.63 ± 0.32	90.13 ± 3.85	88.62 ± 8.01	95.67 ± 2.97	95.67 ± 2.97	95.67 ± 2.97
H3K36me3_TDH_other	77.00 ± 2.83	93.50 ± 2.12	79.00 ± 15.56	91.50 ± 0.71	70.50 ± 17.68	94.09 ± 2.28	94.09 ± 2.28	88.74 ± 5.28
H3K4me1_TDH_BP	78.63 ± 1.69	89.10 ± 2.37	87.02 ± 2.18	85.20 ± 1.68	89.43 ± 1.56	88.39 ± 1.63	88.79 ± 1.07	88.39 ± 1.63
H3K4me3_PGP_immune	59.73 ± 5.00	71.79 ± 3.04	67.76 ± 3.35	70.11 ± 1.27	72.39 ± 0.45	70.68 ± 2.95	72.43 ± 1.89	73.55 ± 0.18
H3K4me3_TDH_ENCODE	92.33 ± 1.83	94.78 ± 1.04	95.37 ± 1.74	97.05 ± 1.25	93.89 ± 2.79	94.60 ± 1.62	94.94 ± 1.14	94.94 ± 1.14
H3K4me3_TDH_immune	68.85 ± 1.39	71.79 ± 3.04	67.76 ± 3.35	70.11 ± 1.27	80.91 ± 3.16	80.61 ± 1.50	81.42 ± 1.50	81.98 ± 1.29
H3K4me3_TDH_other	85.15 ± 4.73	71.79 ± 3.04	67.76 ± 3.35	70.11 ± 1.27	91.83 ± 0.65	89.74 ± 1.76	90.16 ± 1.39	89.90 ± 1.99
H3K4me3_XJ_immune	67.13 ± 0.89	69.44 ± 4.29	71.89 ± 2.29	74.00 ± 3.64	71.98 ± 4.33	70.98 ± 2.36	72.29 ± 3.41	75.06 ± 2.40
H3K9me3_TDH_BP	48.94 ± 1.50	84.29 ± 1.75	85.94 ± 0.11	79.39 ± 6.81	84.19 ± 2.81	86.16 ± 1.43	85.04 ± 1.62	86.08 ± 2.48

6 Paper: Interval Regression Review

Regression models are widely used in real-world applications. However, due to some reasons (errors in measurement, data collection procedures, etc), the target value is *not* a single value but rather an interval of acceptable values. For example, in economic forecasting [Abbara and Zevallos, 2018], environmental studies [Ellison et al., 2014], biomedical [Nguyen and Hocking, 2025a], precise measurements at a specific time period are unreliable, leading researchers to use the confidence interval of measurements over time. Similarly, in engineering [Levitin et al., 2011], multiple measurements of a single statistic (temperature, velocity, etc) are taken, and the confidence interval is considered as the target. In survival analysis [Fazil Baksh et al., 2006], the exact survival time of a patient is often uncertain, it is typically known that the patient has survived for at least a certain number of years. These uncertainties motivate the study of Interval Regression.

The term "regression" generally refers to standard regression, in which each instance corresponds to a single target value. Common models for this task include linear regression [Legendre, 1805], k-nearest neighbors (kNN) [Fix and Hodges, 1985], decision trees [Breiman et al., 1984], random forests [Breiman, 2001], and multilayer perceptrons [Hornik et al., 1989]. This study focuses on Interval Regression setting, where each instance is associated with an interval of acceptable values, called *target interval*. There are 4 types of target intervals (y_l, y_u) based on the end values: uncensored ($-\infty < y_l = y_u < \infty$), right-censored ($-\infty < y_l < y_u = \infty$), left-censored ($-\infty = y_l < y_u < \infty$), and interval-censored ($-\infty < y_l < y_u < \infty$). Many studies contained "interval-valued" or "interval regression" in the title, such as [Tanaka and Lee, 1998, Jeng et al., 2003, Billard and Diday, 2000, Hao, 2009, Hamed et al., 2012, Wang et al., 2015, de A. Lima Neto and de A.T. de Carvalho, 2010, Roque et al., 2007, Ak et al., 2015], the setting of these studies differs from this work since they focus on predicting an interval of real values for each instance, i.e., prediction $\hat{y} = (\hat{y}_l, \hat{y}_u) \in \mathbf{R}^2$ where $\hat{y}_l \leq \hat{y}_u$ or using interval-valued inputs instead of outputs. Instead, in this study's Interval Regression setting, the model predicts a single value from a set of fixed value inputs for each instance, refer to Table 1.

model name	left-censored	right-censored	interval-censored	scalar output	base learner	citation
standard linear	✗	✗	✗	✓	Linear	[Legendre, 1805]
standard knn	✗	✗	✗	✓	kNN	[Fix and Hodges, 1985]
standard tree	✗	✗	✗	✓	Tree	[Breiman et al., 1984]
standard forest	✗	✗	✗	✓	Ensemble trees	[Breiman, 2001]
standard mlp	✗	✗	✗	✓	MLP	[Hornik et al., 1989]
interval-linear	✓	✓	✓	✗	Linear	[Wang et al., 2015]
quadratic	✓	✓	✓	✗	Linear	[Tanaka and Lee, 1998]
svm	✓	✓	✓	✗	Linear	[Jeng et al., 2003]
aft	✗	✓	✗	✓	Linear	[Wei, 1992]
reg-aft	✗	✓	✗	✓	Linear	[Huang et al., 2006]
middlemost	✗	✗	✓	✓	Linear, MLP	[Cheng et al., 2023]
interval-CART	✗	✗	✓	✓	Tree	[Drouin et al., 2017]
interval-knn	✓	✓	✓	✗	kNN	[Hamed et al., 2012]
linear	✓	✓	✓	✓	Linear	[Rigaill et al., 2013]
mmit	✓	✓	✓	✓	Tree	[Drouin et al., 2017]
aft_xgb	✓	✓	✓	✓	Ensemble trees	[Barnwal et al., 2022]
constant	✓	✓	✓	✓	Constant	baseline
knn	✓	✓	✓	✓	kNN	proposed
mlp	✓	✓	✓	✓	MLP	proposed
mmif	✓	✓	✓	✓	Ensemble trees	proposed

Table 10: List of Regression models with compatible censored target intervals and output setting. Only fully compatible models (match the setting of Table 1) are employed.

In some studies, Interval Regression models are trained by converting the problem into Standard Regression: instead of using intervals as targets, the target interval is represented by discrete points, which are then treated as Standard Regression targets. Using the same notations in Table 1 for $\mathbf{x}$ as the features set and (y_l, y_u) as the target interval, Drouin et al. [2017] transform the instance $\big(\mathbf{x}, (y_l, y_u)\big)$ into two points, $(\mathbf{x}, y_l)$ and $(\mathbf{x}, y_u)$, called Interval-CART model whereas Cheng et al. [2023] convert it into its midpoint, $(\mathbf{x}, \frac{y_l+y_u}{2})$, called Middlemost – Approach 1 and 2 in Figure 2. While these approaches are reasonable, it overlooks the interval information by representing intervals by finite values and fails to handle left-censored or right-censored intervals, resulting in wasted data. So the performance of these models is worse compared to the models specifically designed for interval targets, see the results in [Drouin et al., 2017, Cheng et al., 2023] and Figure 2 example.

Interval Regression encompasses Survival Analysis as a special case, where Accelerated Failure

Time (AFT) models are specifically designed for survival data. Survival analysis involves predicting the time until an event occurs, using censored data where the exact event time is unknown. However, a limitation of these models is that they can only handle uncensored and right-censored intervals, which makes them unsuitable for the general setting of Interval Regression.

To date, existing Interval Regression models for this setting include Linear [Rigaill et al., 2013, Cheng et al., 2023, Nguyen and Hocking, 2025b], Tree-based [Drouin et al., 2017, Barnwal et al., 2022] and Accelerated Failure Time (AFT) models family [Buckley and James, 1979, Wei, 1992, Huang et al., 2006, Cai et al., 2009].

6.1 Interval Regression Models

This section details Interval Regression models, including existing and proposed ones. See Table 8 for all of the employed models in this study. The notation used hereafter follows Table 1, where $\mathbf{x} \in \mathbf{R}^m$ is the model input (a vector of m features), $(y_l, y_u) \in \mathbf{R}^2$ where $y_l \leq y_u$ is the target interval, and $\hat{y} \in \mathbf{R}$ is the model prediction.

6.1.1 Existing models

Max Margin Interval Regression (linear). This linear model was introduced by Rigaill et al. [2013], and its extended theoretical foundations were later studied by Cheng et al. [2023]. The model formula is given by:

$$\hat{y} = \mathbf{x} \cdot \beta + \beta_0 \tag{11}$$

where $\beta \in \mathbf{R}^m$ and $\beta_0 \in \mathbf{R}$ are the parameters. To estimate these parameters, the authors extended the Standard Regression loss function — the Squared Error (SE) — into a generalized form called the *Hinge Squared Error*. The error value between prediction $\hat{y}$ and target interval (y_l, y_u) is defined

as:

$$l\Big(\hat{y}, (y_l, y_u)\Big) = \begin{cases} (y_l - \hat{y} + \epsilon)^2, & \text{if } \hat{y} \leq y_l + \epsilon \\ 0, & \text{if } y_l + \epsilon < \hat{y} < y_u - \epsilon \\ (\hat{y} - y_u + \epsilon)^2, & \text{if } \hat{y} \geq y_u - \epsilon \end{cases}$$

where $\epsilon \geq 0$ is the margin length. This error becomes SE when $y_l + \epsilon = y_u - \epsilon$. Cheng et al. [2023] proposed a model called *Regression with Interval Targets* (RIT), which is the same as model (11), but uses *Hinge Absolute Error* instead:

$$l\Big(\hat{y}, (y_l, y_u)\Big) = \begin{cases} |y_l - \hat{y} + \epsilon|, & \text{if } \hat{y} \leq y_l + \epsilon \\ 0, & \text{if } y_l + \epsilon < \hat{y} < y_u - \epsilon \\ |\hat{y} - y_u + \epsilon|, & \text{if } \hat{y} \geq y_u - \epsilon \end{cases}$$

The general Interval Regression loss function can be written in this short form:

$$l\Big(\hat{y}, (y_l, y_u)\Big) = \Big(\text{ReLU}(y_l - \hat{y} + \epsilon)\Big)^p + \Big(\text{ReLU}(\hat{y} - y_u + \epsilon)\Big)^p \tag{12}$$

where $p \in \{1, 2\}$. See Figure 4 for the visual comparison between hinge errors and SE.

To estimate the parameters β and β_0 in (11), apply a gradient descent algorithm with respect to β and β_0 to minimize the sum of Hinge Error (2) over the training set having N instances $\{\mathbf{x}^i, (y_l^i, y_u^i)\}_{i=1}^N$:

$$\sum_{i=1}^{N} l\Big(\hat{y}^i, (y_l^i, y_u^i)\Big) = \sum_{i=1}^{N} l\Big(\mathbf{x^i} \cdot \beta + \beta_0, (y_l^i, y_u^i)\Big)$$

Rigaill et al. [2013] employed the Fast Iterative Shrinkage-Thresholding Algorithm (FISTA) [Beck and Teboulle, 2009], while Cheng et al. [2023] employed stochastic gradient descent (SGD).

Maximum Margin Interval Trees (MMIT). This Tree model was introduced by Drouin et al. [2017] as an alternative nonlinear model to *Max Margin Interval Regression* [Rigaill et al., 2013].

The prediction is given by:

$$\hat{y} = \mathcal{T}(\mathbf{x})$$

where $\mathcal{T}$ represents the tree model, having the same structure as CART (Classification And Regression Tree) [Breiman et al., 1984]. The difference is in the regression value for each leaf which is a set of S instances $\{\mathbf{x}^i, (y_l^i, y_u^i)\}_{i=1}^S$: instead of taking the mean of all targets like CART does, it chooses a constant value c that minimizes the mean Hinge Error (2) between this constant and the target intervals:

$$\hat{y} = \underset{c}{\operatorname{argmin}} \frac{1}{S} \sum_{i=1}^{S} l\big(c, (y_l^i, y_u^i)\big), \quad c \in \mathbf{R}$$

A heuristic algorithm is used to speed up the estimation of the mean value for each leaf by considering a finite set of candidates, rather than solving for the optimal c:

$$\hat{y} = \underset{c}{\operatorname{argmin}} \frac{1}{S} \sum_{i=1}^{S} l\big(c, (y_l^i, y_u^i)\big), \quad c \in \{y_l^i, y_u^i, \frac{y_l^i + y_u^i}{2} | y_l^i > -\infty, y_u^i < \infty\}_{i=1}^S \tag{13}$$

The training procedure follows the same operations as in CART to minimize the sum of Hinge Error (2) over the training set. having N instances $\{\mathbf{x}^i, (y_l^i, y_u^i)\}_{i=1}^N$:

$$\sum_{i=1}^{N} l\Big(\hat{y}^i, (y_l^i, y_u^i)\Big) = \sum_{i=1}^{N} l\Big(\mathcal{T}(\mathbf{x}^i), (y_l^i, y_u^i)\Big)$$

AFT Model in XGBoost (aft_xgb).

$$y_p = \mathbf{T}(\mathbf{x}),$$

Unlike classical AFT models, which can handle only non-negative uncensored and right-censored intervals, the AFT model in XGBoost [Barnwal et al., 2022] can accommodate all non-negative intervals. To extend its applicability to real-valued intervals, they applied an exponential transformation to the original interval. After making a prediction in the exponential space, the result is mapped back to the original scale using the logarithm transformation. The model makes predictions

as follows:

$$\hat{y} = \log \mathbf{T}(\mathbf{x})$$

where $\mathbf{T}$ represents the ensemble of trees in XGBoost. Training this model is based on minimizing the negative log-likelihood function (loss function), which incorporates all training instances:

$$l_{AFT}\Big(\hat{y}, (y_l, y_u)\Big) = \begin{cases} -\log\big(f_Z(\frac{\log y_u - \hat{y}}{\sigma}) \times \frac{1}{\sigma y_u}\big) & \text{if } y_l = y_u \\ -\log\Big[F_Z\big(\frac{\log y_u - \hat{y}}{\sigma}\big) - F_Z\big(\frac{\log y_l - \hat{y}}{\sigma}\big)\Big] & \text{if } y_l < y_u \end{cases} \tag{14}$$

where $\sigma > 0$ is the scale parameter. The Probability Density Function $f_Z(z)$ and Cumulative Distribution Function $F_Z(z)$ depends on the chosen distribution:

- normal: $f_Z(z) = \frac{\exp(-z^2/2)}{\sqrt{2\pi}}, \quad F_Z(z) = \frac{1}{2}\big(1 + \text{erf}(\frac{z}{\sqrt{2}})\big)$
- logistic: $f_Z(z) = \frac{e^z}{(1+e^z)^2}, \quad F_Z(z) = \frac{e^z}{1+e^z}$
- extreme: $f_Z(z) = e^z e^{-\exp z}, \quad F_Z(z) = 1 - e^{-\exp z}$

Figure 3 visualizes the loss function for each distributions, which can be compared to Figure 4 to observe their similarities. The model is trained using the gradient boosting algorithm [Friedman, 2001] to minimize the loss value (8) over the training dataset given the distribution and the scale parameter σ.

$$\sum_{i=1}^{N} l_{AFT}\Big(\hat{y}^i, (y_l^i, y_u^i)\Big) = \sum_{i=1}^{N} -\log\Big[F_Z\big(\frac{\log y_u^i - \log \mathbf{T}(\mathbf{x}^i)}{\sigma}\big) - F_Z\big(\frac{\log y_l^i - \log \mathbf{T}(\mathbf{x}^i)}{\sigma}\big)\Big]$$

6.1.2 Proposed Models

To provide a more comprehensive comparison, this study proposes several alternative machine learning models, such as KNN, Multilayer Perceptron and Forest.

dataset	features	instances	features pattern/quality			
			linear	nonlinear	noisy	excessive
auto93	62	82	✓			✓
autohorse	59	159	✓			✓
autompg	25	392	✓			
autoprice	15	159		✓		
baskball	4	96	✓			
bodyfat	14	252	✓			
breasttumor	40	286	✓			
cholesterol	26	299	✓		✓	
cleveland	26	299	✓			
cloud	10	108		✓		
cpu	36	209		✓		
echomonths	13	106	✓			
elusage	13	55	✓			
fishcatch	15	158	✓			
fruitfly	8	125	✓		✓	
housing	14	506		✓		
lowbwt	23	189		✓		
machine.cpu	6	209		✓		
mbagrade	3	61		✓	✓	
meta	54	264	✓			✓
pbc	29	276	✓			
pollution	15	60		✓		
pwlinear	10	200		✓		
pyrim	27	74		✓	✓	
sensory	36	576	✓		✓	
servo	19	167		✓		
sleep	7	51	✓			
stock	9	950		✓		
strike	23	625	✓			
triazines	60	186		✓		✓
veteran	13	137	✓			
vineyard	3	52	✓			
wisconsin	32	194		✓		
simulated.abs	20	200		✓	✓	
simulated.linear	20	200	✓		✓	
simulated.sin	20	200		✓	✓	
pharynx	218	195		✓		✓
lymphoma.mkatayama	258	41		✓	✓	✓
lymphoma.tdh	258	23	✓		✓	✓

Table 11: Datasets list for Interval Regression study.

Note: Linear indicates that the relationship between features and the target variable is likely linear, while nonlinear indicates otherwise. Noisy refers to datasets where more than half of the features are irrelevant. Excessive describes datasets with more features than instances.

kNN. kNN is a classical model for Standard Regression, which motivates its consideration for Interval Regression. The operation follows the same steps as the standard kNN regression model

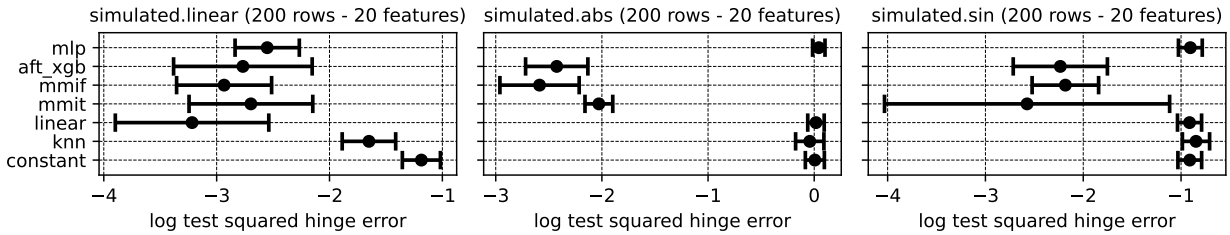


Figure 21: The mean and standard deviation of the test squared hinge errors in simulated datasets

[Fix and Hodges, 1985], with the difference being how the regression value is determined from the k nearest neighbors. We propose utilizing the MMIT regression function. Specifically, we treat the set of k nearest neighbors as a single leaf in a MMIT, where the regression value is a constant that minimizes the mean Hinge Error (2) between this constant and the k target intervals. The prediction $\hat{y}$ based on the target intervals of the k nearest neighbors, denoted as $\{(y_l^i, y_u^i)\}_{i=1}^k$, is formulated as:

$$\hat{y} = \underset{c}{\operatorname{argmin}} \frac{1}{k} \sum_{i=1}^{k} l\big(c, (y_l^i, y_u^i)\big), \quad c \in \{y_l^i, y_u^i, \frac{y_l^i + y_u^i}{2} | y_l^i > -\infty, y_u^i < \infty\}_{i=1}^k$$

Multilayer Perceptron (MLP). Since MLP is a commonly used neural network in Standard Regression models [Hornik et al., 1989], its application in Interval Regression is worth considering. The training process of this model is the same as in the linear model *Max Margin Interval Regression*.

Maximum Margin Interval Forest (MMIF). The idea of using ensembles of MMITs was mentioned in Chapter 6 (Discussion and Conclusions) of [Drouin et al., 2017]. Since MMIT is the Interval Regressor Tree, its extension to construct a Forest is straightforward. The operation of MMIF follows the same principles as the standard Random Forest regressor [Breiman, 2001], with the key distinction being that MMITs are used instead of CART.

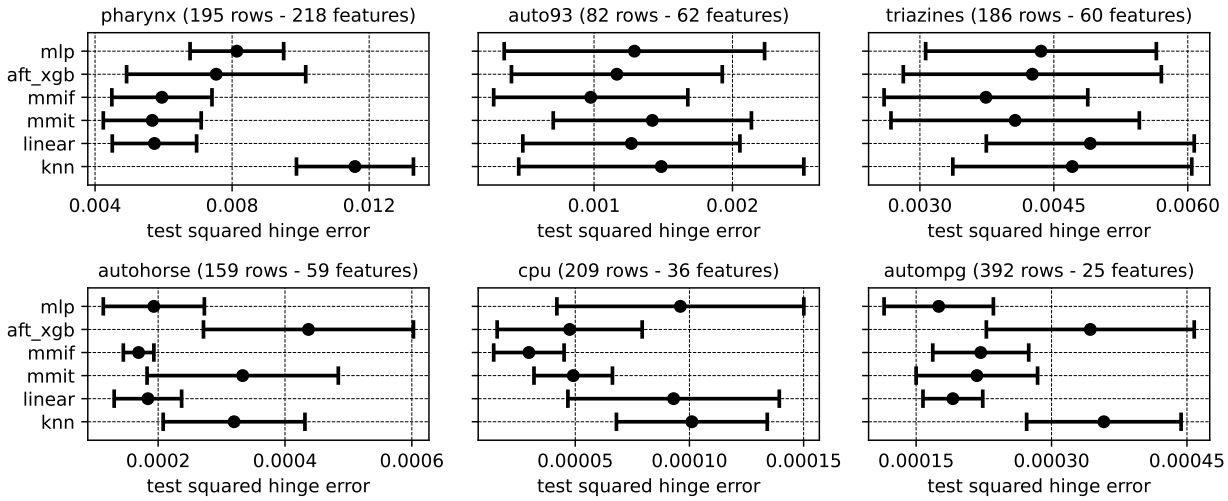


Figure 22: The mean and standard deviation of the test squared hinge errors in real datasets

6.2 Experiments and Results

Experimental Datasets. To ensure a comprehensive comparison across different scenarios, we evaluate all the models on a diverse set of datasets. Specifically, we use 36 real-world datasets from the UCI repository [Lichman, 2013] along with 3 simulated datasets - identical to those in [Drouin et al., 2017]. Each simulated dataset consists of 200 instances with 20 features, where 19 are noise and one is the true feature x. The target intervals are defined based on the linear function of x, $\sin(x)$, or the absolute value of x. The targets in all datasets were transformed into target intervals as follows:

$$y_i \rightarrow (y_i - \varepsilon, y_i + \varepsilon) \qquad \text{where } \varepsilon \in \mathcal{N}(0, \frac{|y_i| + 10^{-6}}{10})$$

Then about 20% of the interval bounds was removed to generate open interval (left or right censored). All datasets are in Table 4.2 and are accessible here: datasets.

Evaluation Metrics. Each dataset is divided into 5 similar sized folds. Each fold is used as the test set, while the remaining 4 folds are combined to form the train set. For each train/test pair, the model is trained on the train set and used to make predictions on the test set. The test set is denoted

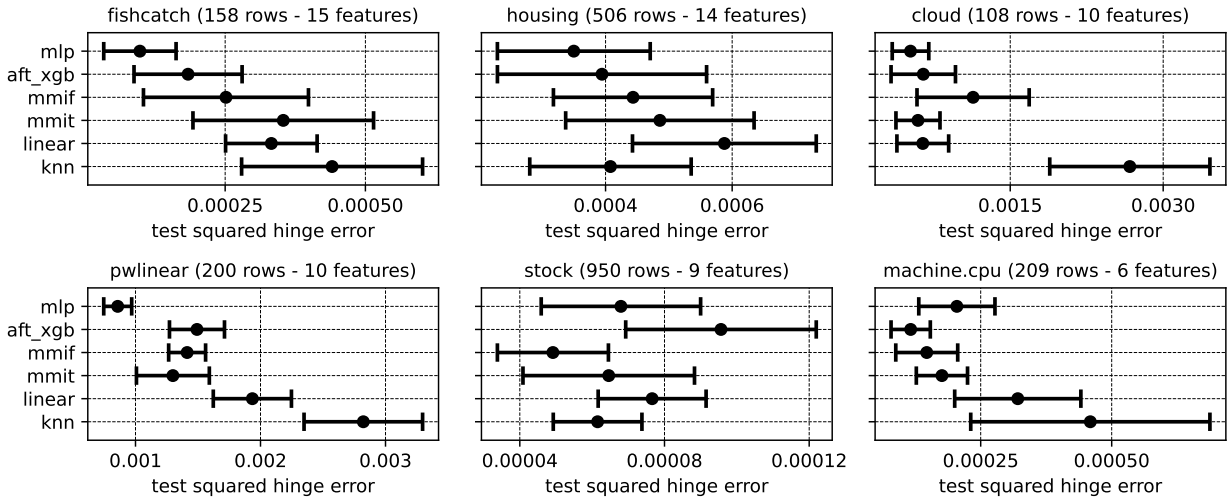


Figure 23: The mean and standard deviation of the test squared hinge errors in small datasets

as having M instances with corresponding target intervals $\{(y_l^i, y_u^i)\}_{i=1}^M$, and the predictions set is $\{\hat{y}^i\}_{i=1}^M$. The Mean Squared Hinge Error is given by:

$$\frac{1}{M}\sum_{i=1}^{M}\big(\text{ReLU}(y_l^i - \hat{y}^i)\big)^2 + \big(\text{ReLU}(\hat{y}^i - y_u^i)\big)^2 \tag{15}$$

which is equivalent to using Hinge Error (2) with $p = 2$ and $\epsilon = 0$. Each train/test pair produces one error values. From these 5 error values, the mean and standard deviation are computed. Two metrics are then considered: **performance** and **consistency**. A smaller mean error indicates better performance, while a smaller standard deviation indicates better consistency.

6.2.1 Model Configuration

Except for the AFT model in XGBoost, the Squared Hinge Error is used as the loss function for both the cross-validation process and model training. The configuration for each model below is applied to a single train/test set pair.

Constant (Featureless). This model predicts a single value using only training target intervals. The prediction is a constant that minimizes the Mean Squared Hinge Error (15) between this constant and the set of target intervals.

Linear. The model *Max Margin Interval Regression* model is implemented with L1 regularization. The regularization parameter starts at 0.001, geometrically increasing by a factor of 1.2 until no features remain, with cross-validation (`cv` = 5) on the train set to determine the optimal L1 value.

MMIT. A cross-validation (`cv` = 5) was used to select the optimal hyperparameters, including `max_depth` $\{0, 1, 5, 10, 20, \infty\}$ and `min_sample` $\{0, 1, 2, 4, 8, 16, 20\}$.

AFT model in XGBoost. A cross-validation (`cv` = 5) was performed to select the optimal hyperparameters, following the same grid search in Barnwal et al. [2022]:

- `learning_rate`: 0.001, 0.01, 0.1, 1.0
- `max_depth`: 2, 3, 4, 5, 6, 7, 8, 9, 10
- `min_child_weight`: 0.001, 0.1, 1.0, 10.0, 100.0
- `reg_alpha`: 0.001, 0.01, 0.1, 1.0, 10.0, 100.0
- `reg_lambda`: 0.001, 0.01, 0.1, 1.0, 10.0, 100.0
- `aft_loss_distribution_scale`: 0.5, 0.8, 1.1, 1.4, 1.7, 2.0

kNN. The Euclidean distance metric is applied to normalized features (mean 0, standard deviation 1) to determine the nearest neighbors. A cross-validation (`cv` = 5) is used to select the optimal value of K. The candidate values for K range from 1 to $\lceil \sqrt{n} \rceil$, where n is the number of train instances.

	Performance							Consistency						
	1st	2nd	3rd	4th	5th	6th	7th	1st	2nd	3rd	4th	5th	6th	7th
constant	2	2	3	1	4	7	20	4	0	2	3	3	8	19
knn	4	4	2	6	6	16	1	4	3	3	7	7	12	3
linear	3	11	6	8	5	5	1	7	7	8	7	7	3	0
mmit	2	10	9	3	9	4	2	2	13	4	8	8	2	2
mmif	15	5	8	7	4	0	0	12	9	9	4	3	2	0
aft_xgb	4	4	3	4	7	5	12	4	1	7	2	3	10	12
mlp	9	3	8	10	4	2	3	6	6	6	8	8	2	3

Table 12: The comparison of the performance and consistency of 7 interval regression models

MLP. A cross-validation (`cv` = 5) is used to determine the optimal hyperparameters:

- `num_layer`: {1, 2}
- `hidden_layer_size`: {5, 10, 20}
- `activation`: {ReLU, Sigmoid}

The model is trained using the Adam optimizer with a fixed learning rate of 0.001.

MMIF. MMIF is an ensemble of many MMITs, we chose 100. For each MMIT, two-thirds of the train dataset with one-third of the features are randomly selected for training, while the remaining instances serves as the out-of-bag (OOB) set for validation. A cross-validation (`cv` = 5) is used to select the optimal hyperparameters for each MMIT including `max_depth` {2, 5, 10, 15, 20, 25} and `min_split_samples` {2, 5, 10, 20, 50}. Each MMIT has its own OOB error, which is used to determine the aggregation rule. Let $\mathcal{T}_i$ be an MMIT with OOB error e_i. The weight assigned to $\mathcal{T}_i$ is given by $w_i = \frac{\frac{1}{e_i}}{\sum_{j=1}^{100} \frac{1}{e_j}}$ and the final prediction from MMIF is computed as $\hat{y} = \sum_{i=1}^{100} w_i \mathcal{T}_i(\mathbf{x})$, where $\mathbf{x}$ is the set of features.

6.2.2 Results

For each dataset, 7 models were implemented, each model produces 5 error values. Figure 21 shows the comparison on 3 simulated datasets, where 95% of the features are noise. Figure 22 shows the

comparison on 6 fairly unstable datasets, where the feature-to-instance ratio is high (from 0.06 to 1.12). Figure 23 shows the comparison on 6 fairly stable datasets, where the feature-to-instance ratio is low (from 0.009 to 0.09).

6.3 Discussions and Conclusions

Dataset Variations. To ensure a comprehensive comparison, the quality of the datasets is important. The datasets employed have a variety of characteristics, including high- and low-dimensional data, real-world and synthetic sources, linear and nonlinear patterns, as well as low- and high-noise conditions.

Comprehensive Comparison. The performance and consistency rankings are in Table 12, 1st means the best and 7th means the worst. About the performance, it's clear that MMIF outperforms other models in most cases. Linear, MMIT and MLP are comparable, while Constant, kNN, and the AFT model in XGBoost show the poorest performance. About the consistency, MMIF stands out as the most consistent. This is expected, as MMIF being an ensemble model, relies on a group of MMITs to make the final prediction, which helps improve its stability. Again, Constant, kNN, and the AFT model in XGBoost show the lowest consistency.

No single model is optimal for all scenarios. As shown in Table 12, there is no universally optimal model for every scenario. However, if complexity is not a primary concern, and the goal is to select the model with the best overall performance and consistency, MMIF is likely the first choice. Alternatively, if model complexity and training time are critical factors, MMIT is preferable, as it's less complex than MMIF, require significantly fewer training resources, and still maintain reliable performance.

MMIT vs MMIF. This is a comparison worth considering. MMIF was proposed with the expectation that an ensemble of MMITs would outperform a single MMIT, and the results confirm this

	constant	knn	linear	mmit	mmif	aft_xgb	mlp
auto93	7	6	3	5	1	2	4
autohorse	7	4	2	5	1	6	3
autompg	7	6	2	3	4	5	1
autoprice	7	4	6	2	1	5	3
baskball	6	2	5	3	1	7	4
bodyfat	7	6	2	4	3	5	1
breasttumor	6	1	4	5	2	7	3
cholesterol	3	5	4	6	1	7	2
cleveland	6	4	2	5	3	7	1
cloud	7	6	3	2	5	4	1
cpu	7	6	4	3	1	2	5
echomonths	7	4	2	5	3	6	1
elusage	7	2	1	3	4	5	6
fishcatch	7	6	4	5	3	2	1
fruitfly	1	5	2	6	3	7	4
housing	7	3	6	5	4	2	1
lowbwt	6	5	3	2	1	7	4
lymphoma.mkatayama	1	5	6	3	4	7	2
lymphoma.tdh	3	6	2	4	1	7	5
machine.cpu	7	6	5	3	2	1	4
mbagrade	3	6	5	1	2	4	7
meta	5	1	2	6	4	7	3
pbc	7	5	1	4	2	6	3
pharynx	7	6	2	1	3	4	5
pollution	4	6	3	2	1	7	5
pwlinear	7	6	5	2	3	4	1
pyrim	2	1	6	3	4	5	7
sensory	2	6	3	5	1	7	4
servo	7	6	4	2	5	1	3
simulated.abs	5	4	6	2	1	3	7
simulated.linear	7	6	1	3	2	5	4
simulated.sin	5	7	4	2	1	3	6
sleep	7	1	2	5	3	6	4
stock	7	2	5	3	1	6	4
strike	6	2	4	7	5	1	3
triazines	5	6	7	2	1	3	4
veteran	6	3	2	7	4	5	1
vineyard	7	4	3	6	5	1	2
wisconsin	6	5	4	2	1	7	3

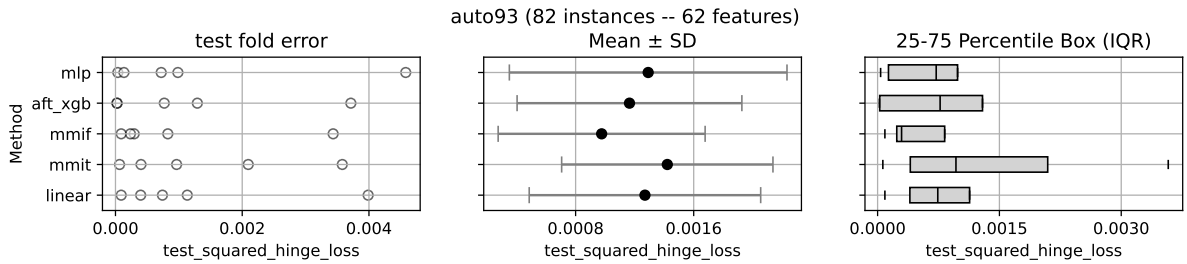
Table 13: The performance ranking of 7 models across 39 datasets.

hypothesis. Specifically, across 39 datasets, MMIF outperformed MMIT in 28 of them. This makes sense: MMIF, being an ensemble of MMITs, is able to reduce overfitting by not relying on a single

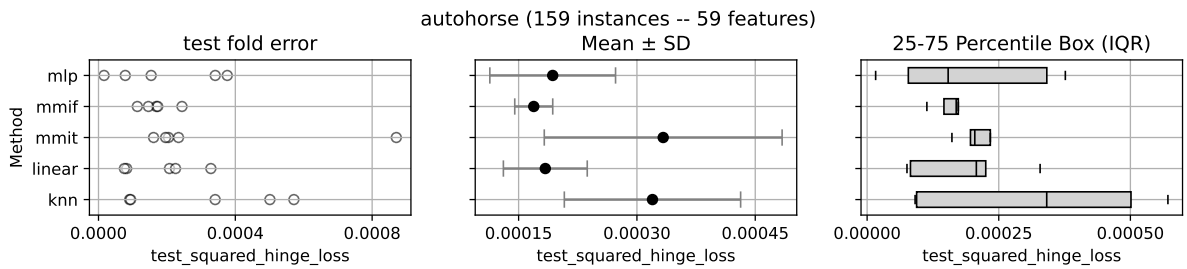
	constant	knn	linear	mmit	mmif	aft_xgb	mlp
auto93	7	6	4	2	1	3	5
autohorse	7	4	2	5	1	6	3
autompg	7	5	1	4	2	6	3
autoprice	7	3	4	2	1	6	5
baskball	5	4	3	2	1	6	7
bodyfat	7	5	1	4	3	6	2
breasttumor	6	1	2	5	3	7	4
cholesterol	5	6	4	7	1	3	2
cleveland	6	5	1	4	3	7	2
cloud	7	6	3	2	5	4	1
cpu	7	4	5	2	1	3	6
echomonths	7	2	5	4	3	6	1
elusage	7	4	1	2	3	5	6
fishcatch	7	6	2	5	4	3	1
fruitfly	3	5	2	7	1	6	4
housing	7	3	4	5	2	6	1
lowbwt	6	4	3	1	2	7	5
lymphoma.mkatayama	1	6	5	2	3	7	4
lymphoma.tdh	3	6	4	2	1	7	5
machine.cpu	7	6	5	2	3	1	4
mbagrade	6	2	5	1	3	4	7
meta	6	1	2	5	4	7	3
pbc	6	2	3	5	4	7	1
pharynx	6	5	1	3	4	7	2
pollution	4	6	3	2	1	7	5
pwlinear	7	6	5	4	2	3	1
pyrim	1	3	6	4	5	2	7
sensory	1	6	3	4	2	7	5
servo	7	5	4	2	6	1	3
simulated.abs	6	7	5	2	1	3	4
simulated.linear	7	6	1	4	2	5	3
simulated.sin	4	7	6	2	1	3	5
sleep	7	4	2	5	1	6	3
stock	7	1	2	5	3	6	4
strike	5	7	6	3	2	1	4
triazines	1	5	3	6	2	7	4
veteran	7	1	4	3	6	5	2
vineyard	7	4	3	6	5	1	2
wisconsin	4	6	1	3	2	7	5

Table 14: The consistency ranking of 7 models across 39 datasets.

MMIT.



Hinge Square Error of 5 models on dataset: auto93



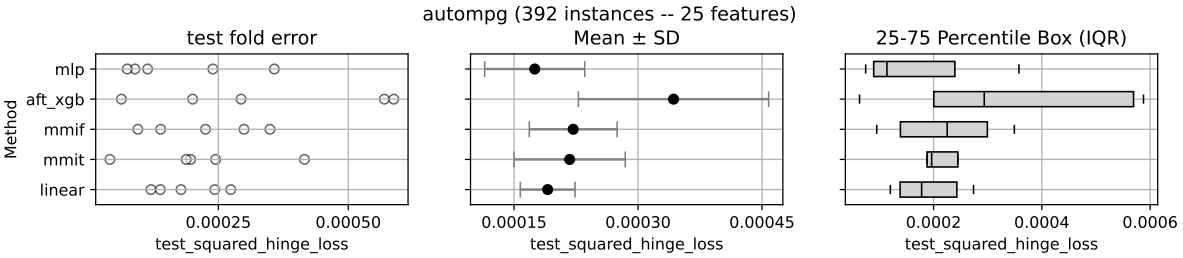
Hinge Square Error of 5 models on dataset: autohorse

Discussion on the AFT Model in XGBoost. The AFT model in XGBoost is the complex models in this study, but its performance and consistency do not justify the complexity. To make the model applicable to a general Interval Regression setting, the target intervals must be transformed using exponential function. So the model becomes highly sensitive when the dataset contains many left-censored target intervals. When predictions are made on the exponential scale, if the predicted value is close to zero, the mapping back to the original scale can result in large negative values.

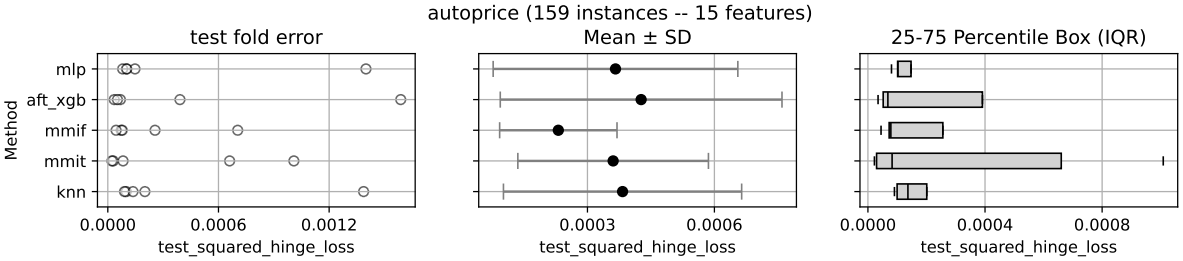
$$(-\infty, y_u) \xrightarrow{\text{exp}} (0, \exp y_u) \xrightarrow{\text{predict}} y_p \approx 0 \xrightarrow{\log} \hat{y} \approx -\infty \xrightarrow{\text{evaluate}} \text{Big test error}$$

The Figure 2.b in [Barnwal et al., 2022] indicates that the AFT model in XGBoost performs among the worst across their six simulated datasets when compared to the Linear and MMIT.

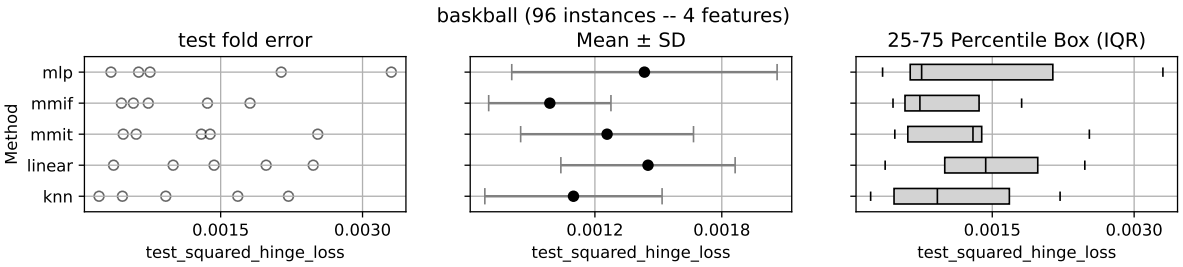
A potential direction for future work is to transform left-censored intervals into interval-censored intervals. It can be done by replacing all lower bounds of $-\infty$ with a finite value. This adjustment aims to reduce the model's sensitivity to left-censored intervals and mitigate the tendency to produce excessively large negative predictions.



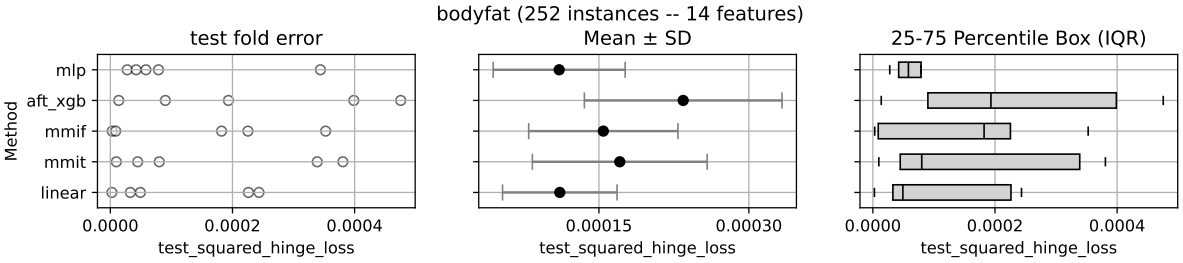
Hinge Square Error of 5 models on dataset: autompg



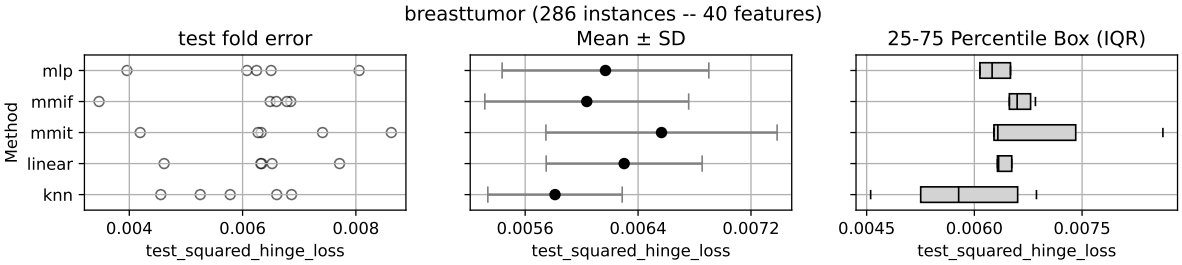
Hinge Square Error of 5 models on dataset: autoprice



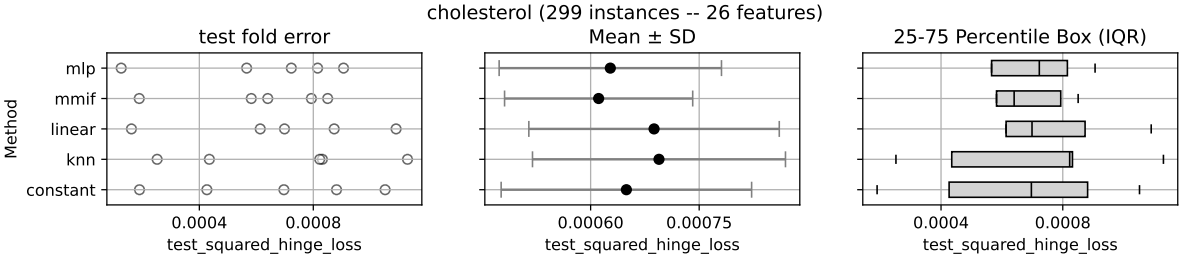
Hinge Square Error of 5 models on dataset: baskball



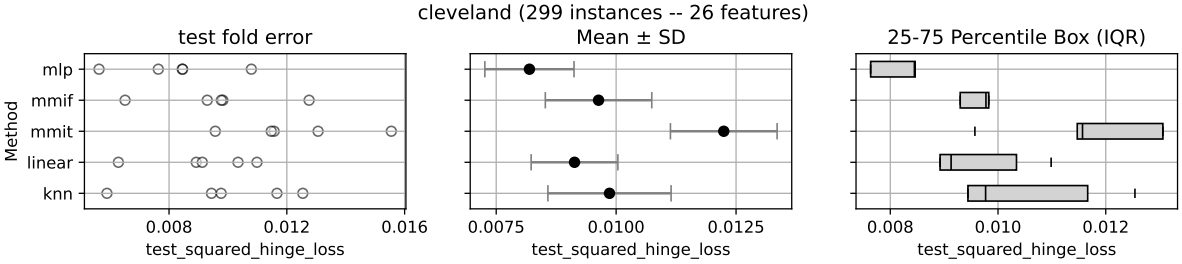
Hinge Square Error of 5 models on dataset: bodyfat



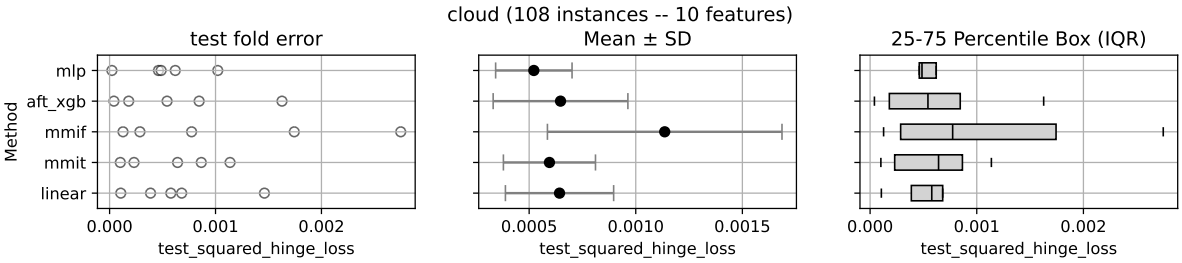
Hinge Square Error of 5 models on dataset: breasttumor



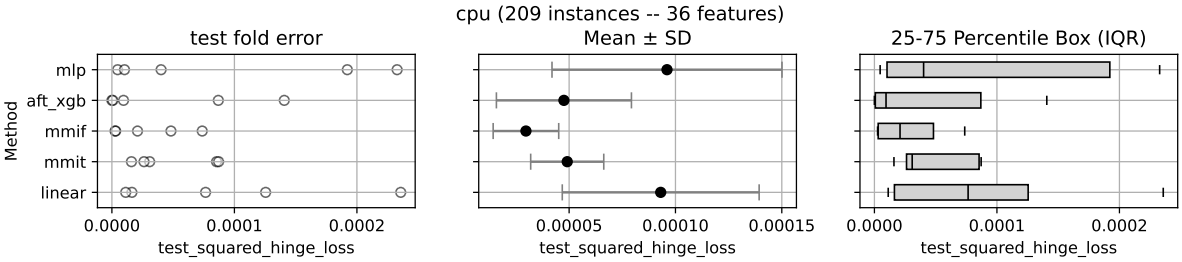
Hinge Square Error of 5 models on dataset: cholesterol



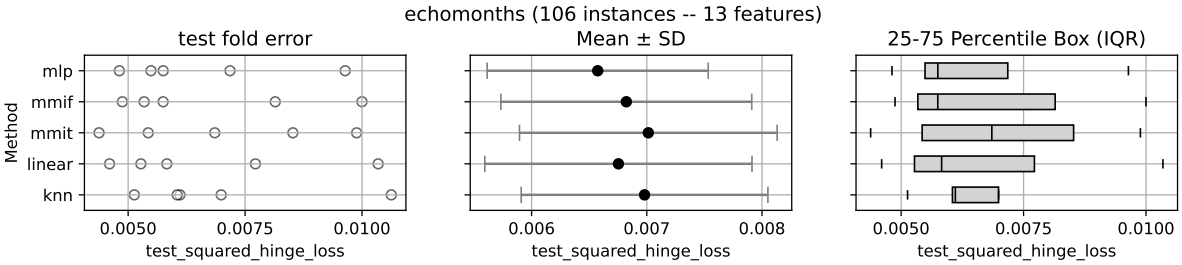
Hinge Square Error of 5 models on dataset: cleveland



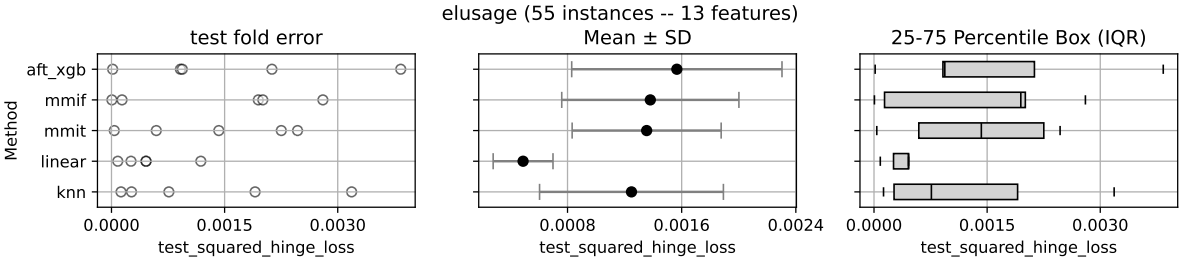
Hinge Square Error of 5 models on dataset: cloud



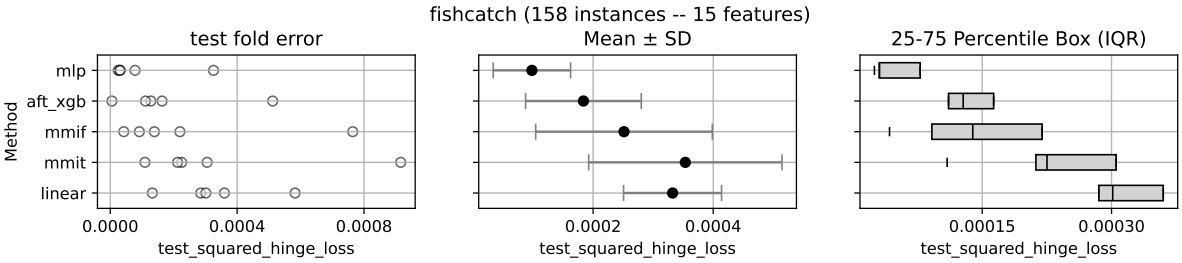
Hinge Square Error of 5 models on dataset: cpu



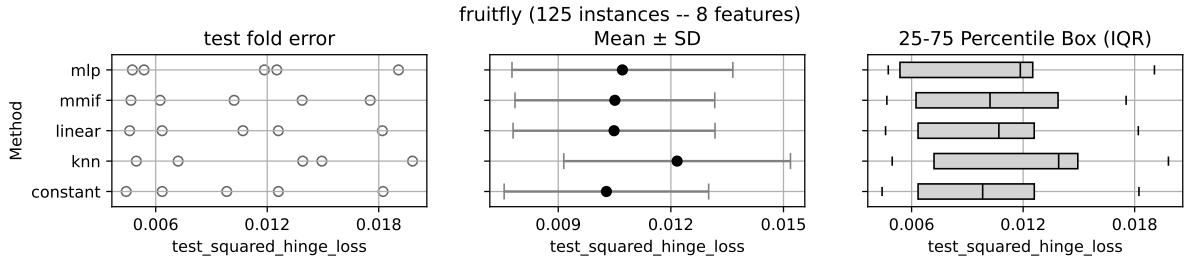
Hinge Square Error of 5 models on dataset: echomonths



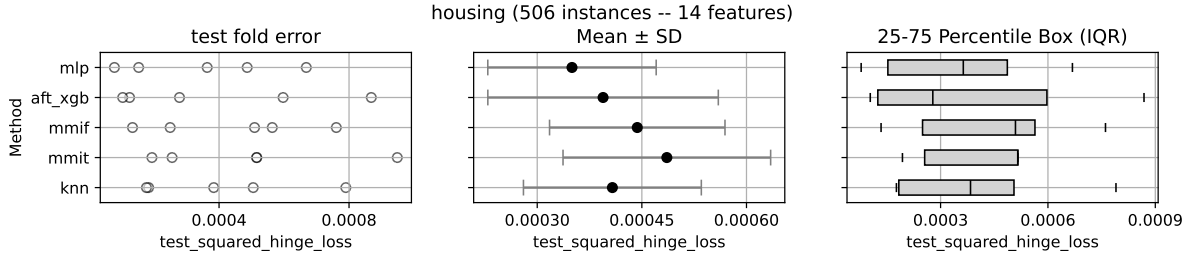
Hinge Square Error of 5 models on dataset: elusage



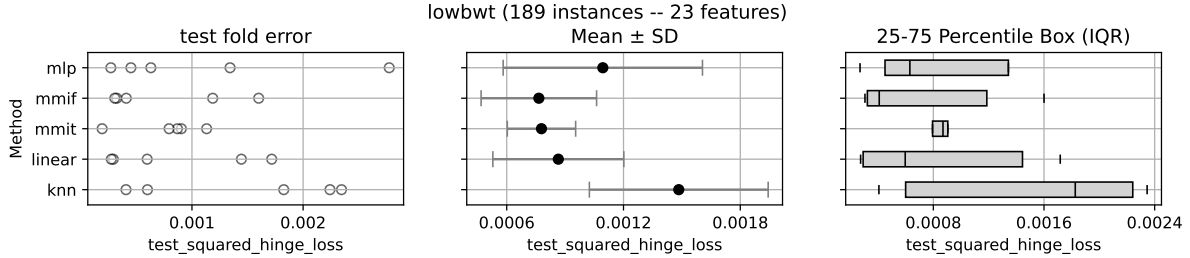
Hinge Square Error of 5 models on dataset: fishcatch



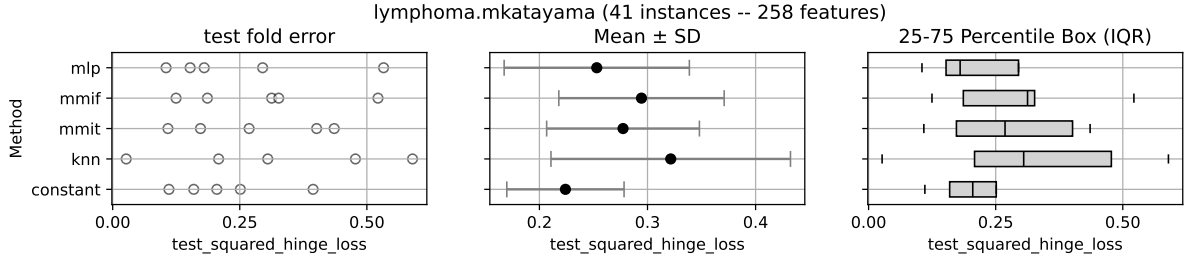
Hinge Square Error of 5 models on dataset: fruitfly



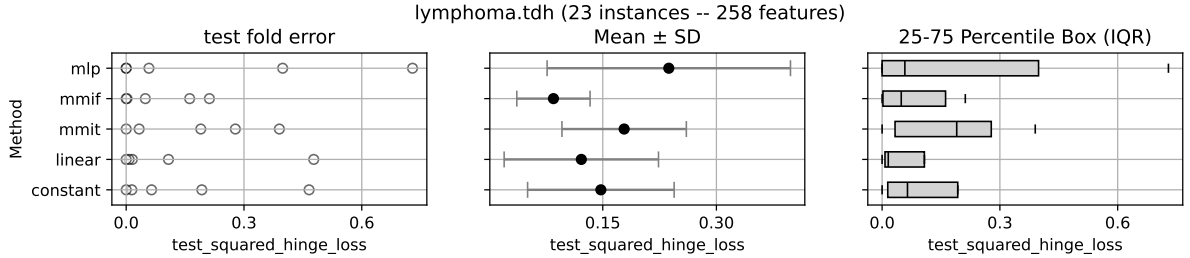
Hinge Square Error of 5 models on dataset: housing



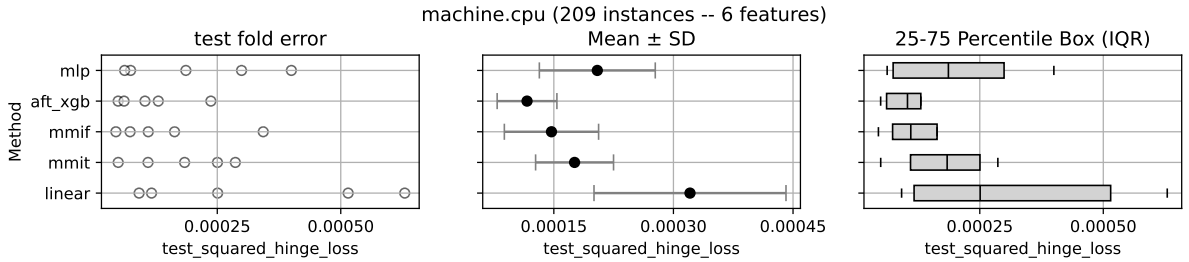
Hinge Square Error of 5 models on dataset: lowbwt



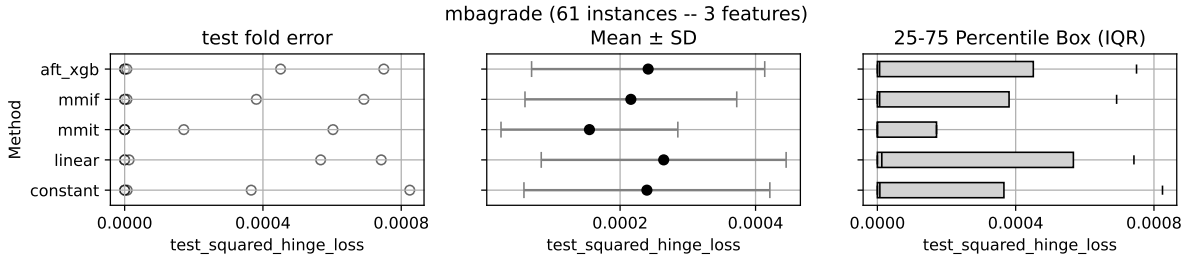
Hinge Square Error of 5 models on dataset: lymphoma.mkatayama



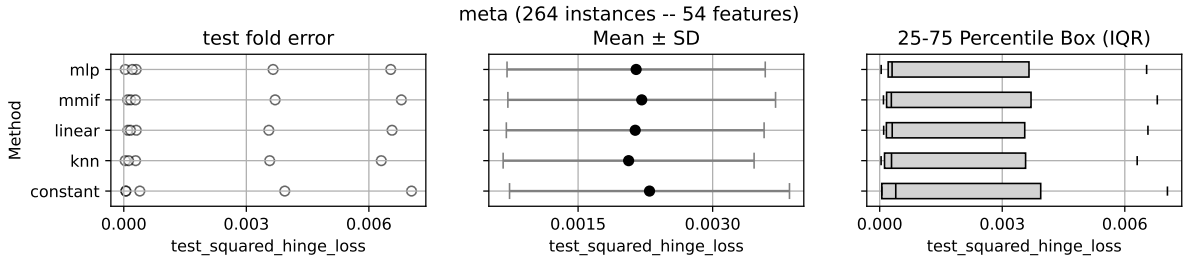
Hinge Square Error of 5 models on dataset: lymphoma.tdh



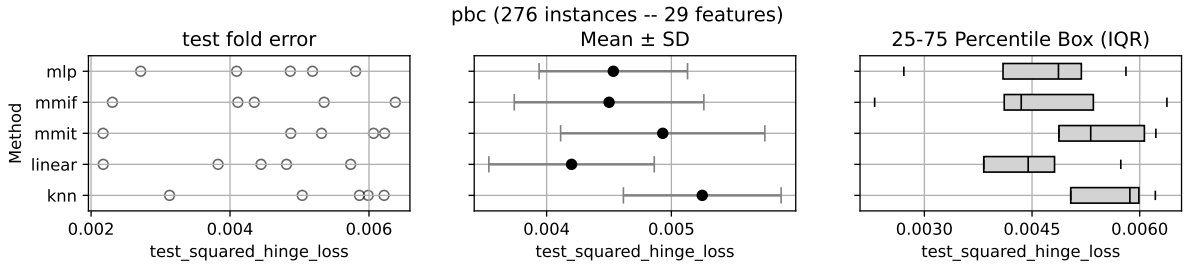
Hinge Square Error of 5 models on dataset: machine.cpu



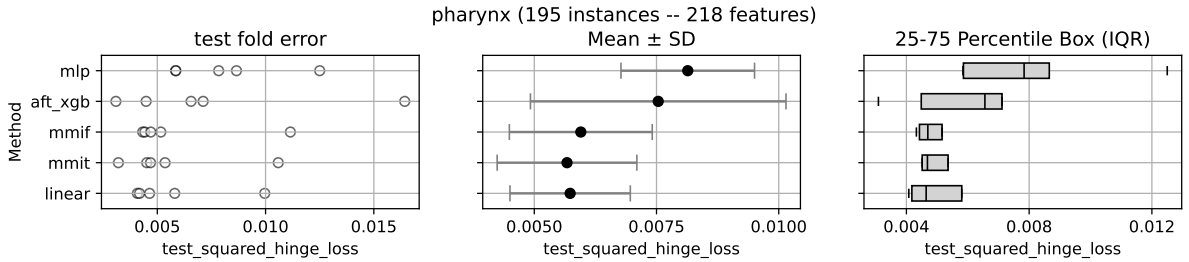
Hinge Square Error of 5 models on dataset: mbagrade



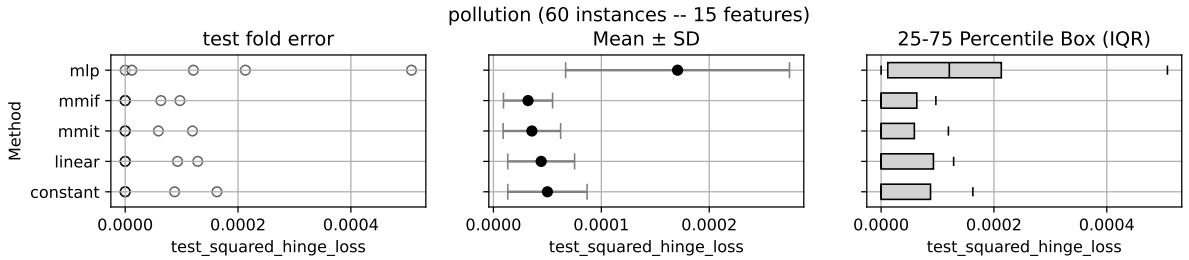
Hinge Square Error of 5 models on dataset: meta



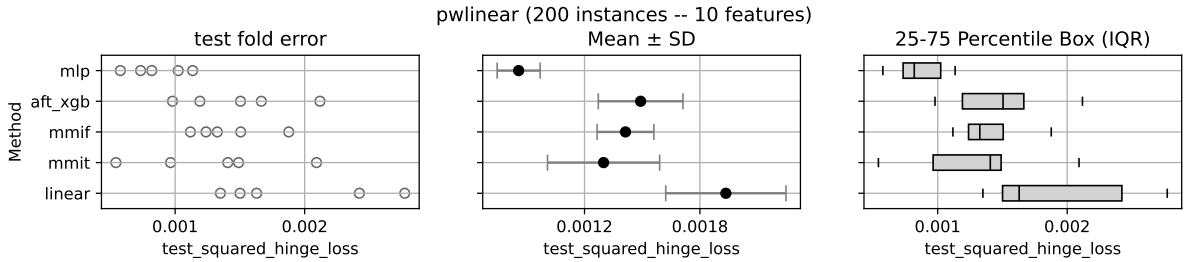
Hinge Square Error of 5 models on dataset: pbc



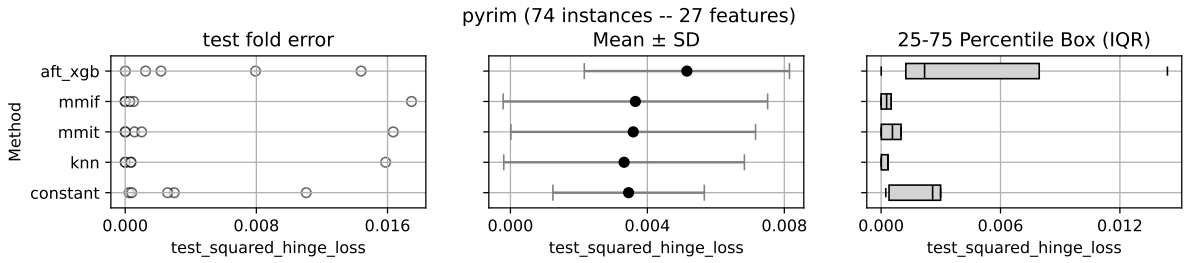
Hinge Square Error of 5 models on dataset: pharynx



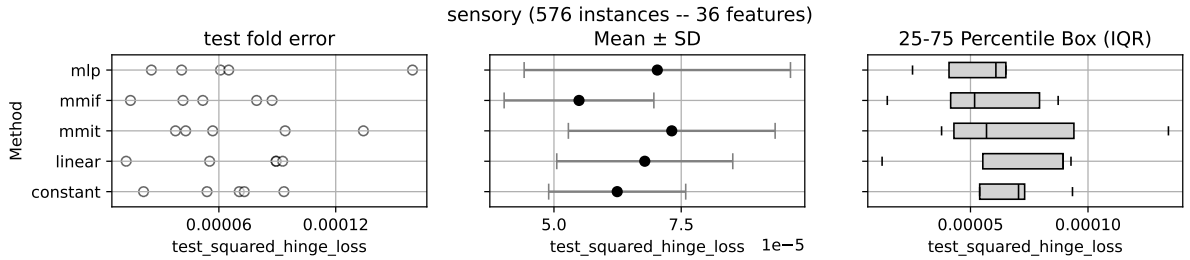
Hinge Square Error of 5 models on dataset: pollution



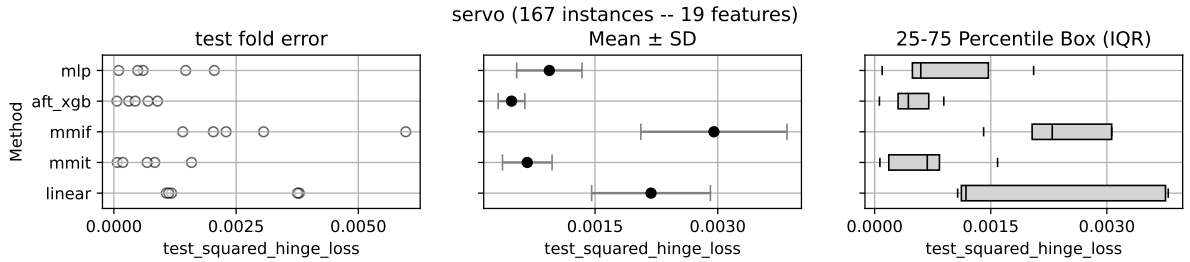
Hinge Square Error of 5 models on dataset: pwlinear



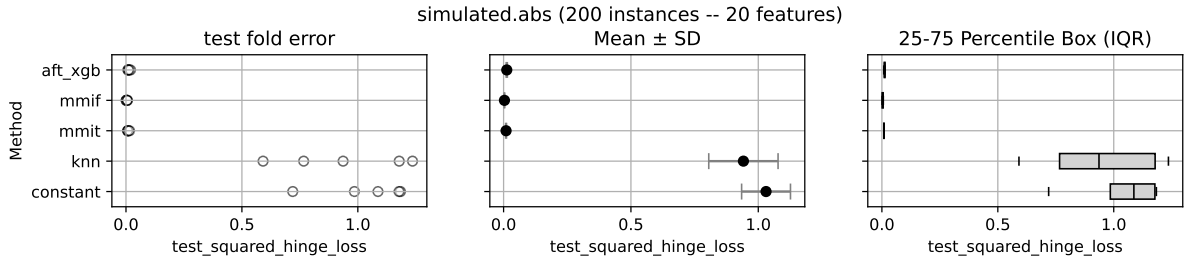
Hinge Square Error of 5 models on dataset: pyrim



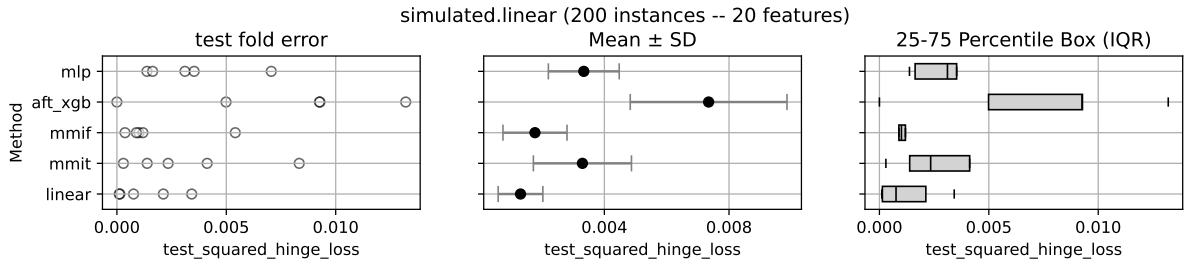
Hinge Square Error of 5 models on dataset: sensory



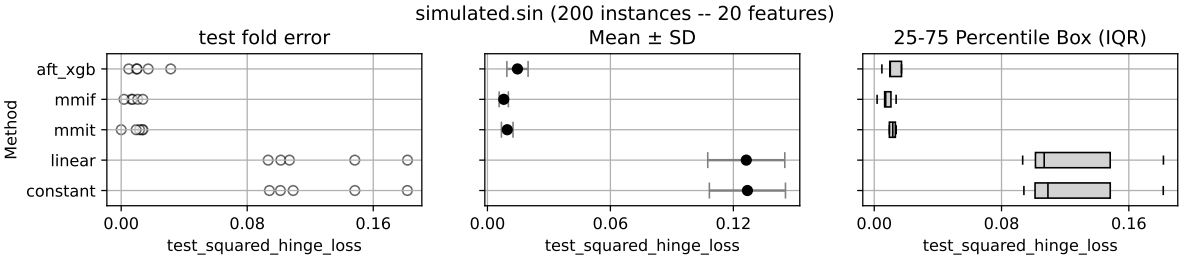
Hinge Square Error of 5 models on dataset: servo



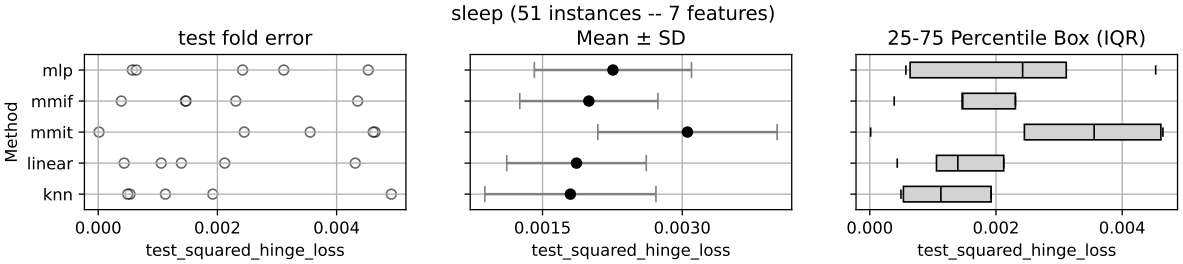
Hinge Square Error of 5 models on dataset: simulated.abs



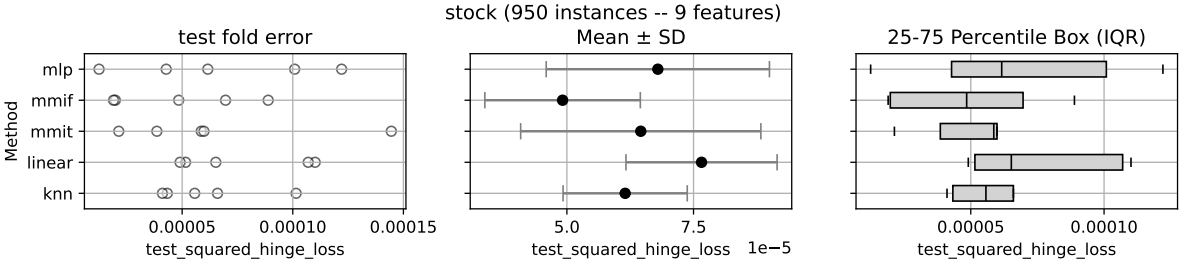
Hinge Square Error of 5 models on dataset: simulated.linear



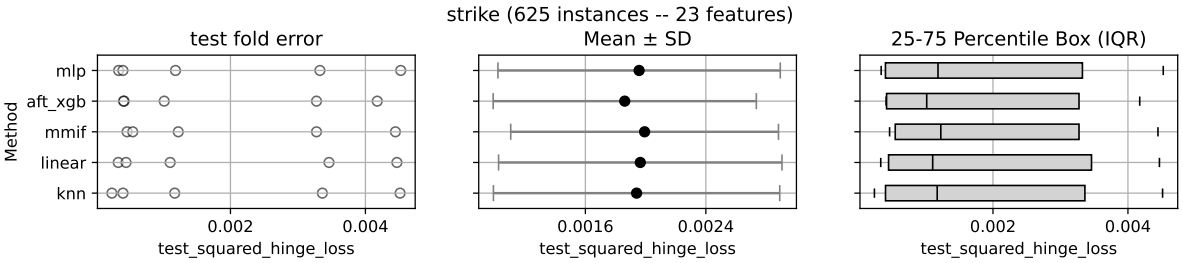
Hinge Square Error of 5 models on dataset: simulated.sin



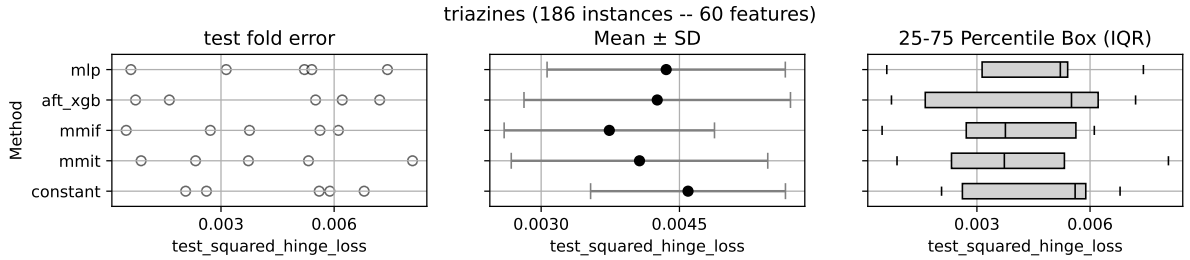
Hinge Square Error of 5 models on dataset: sleep



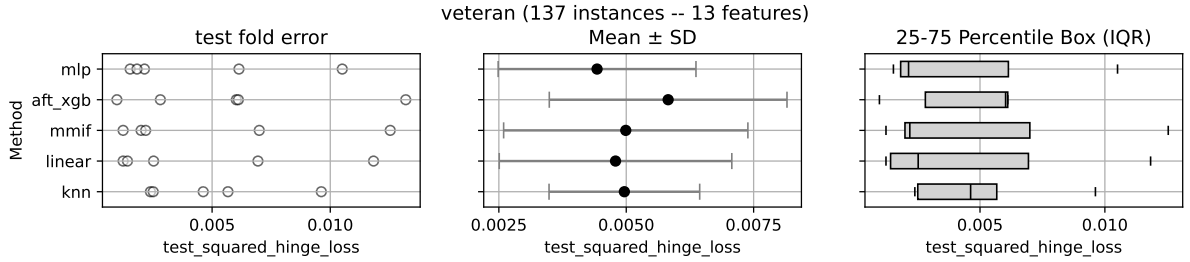
Hinge Square Error of 5 models on dataset: stock



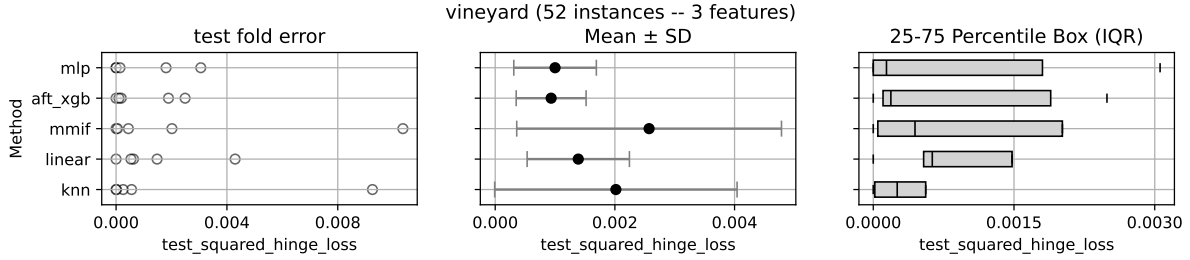
Hinge Square Error of 5 models on dataset: strike



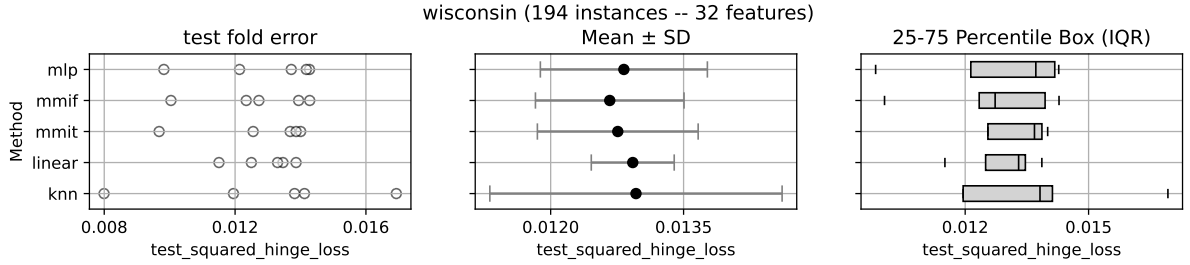
Hinge Square Error of 5 models on dataset: triazines



Hinge Square Error of 5 models on dataset: veteran



Hinge Square Error of 5 models on dataset: vineyard



Hinge Square Error of 5 models on dataset: wisconsin

7 Future Plans

7.1 Optimal Partitioning Penalty Learning using 1D-CNN

Changepoint detection (also known as partitioning or segmentation) plays a crucial role in identifying abrupt structural changes in sequential data across a wide range of domains, including finance [Lattanzi and Leonelli, 2021], healthcare [Muggeo and Adelfio, 2010], network security [Tartakovsky et al., 2013], and environmental science [Reeves et al., 2007]. This broad applicability has motivated the development of numerous detection strategies over the past decades. Existing changepoint detection methods can generally be divided into heuristic approaches and dynamic programming (DP) approaches. Heuristic methods include deep learning–based techniques inspired by classical statistical procedures such as CUSUM [Hinkley, 1971, James et al., 1987], where recent works [Li et al., 2024, Ermshaus et al., 2023] apply multilayer perceptrons (MLP) or k-nearest neighbors (KNN) over sliding windows to classify candidate changepoints. Although effective in some contexts, these methods rely primarily on local neighborhood information rather than the entire sequence and are often unnecessarily complex for univariate data. In contrast, DP-based methods formulate changepoint detection as a global optimization problem and are therefore more suitable when optimal partitioning is desired.

Among DP methods, regularization plays a central role in controlling the number of detected changepoints. Two main strategies are used: specifying a fixed number of changepoints or introducing a penalty term that penalizes changepoint occurrences. Algorithms designed for a fixed number of changepoints include Binary Segmentation and related approaches [Auger and Lawrence, 1989, Bai and Perron, 2003, Killick et al., 2012, Rigaill, 2015]. However, since the true number of changepoints is rarely known in practice, penalty-based approaches are more commonly adopted. Optimal Partitioning (OPART) [Jackson et al., 2005] and its efficient variants—Pruned Exact Linear Time (PELT) [Killick et al., 2012], Functional Pruning Optimal Partitioning (FPOP) [Maidstone et al., 2017], and Labeled Optimal Partitioning (LOPART) [Hocking and Srivastava, 2023]—are widely

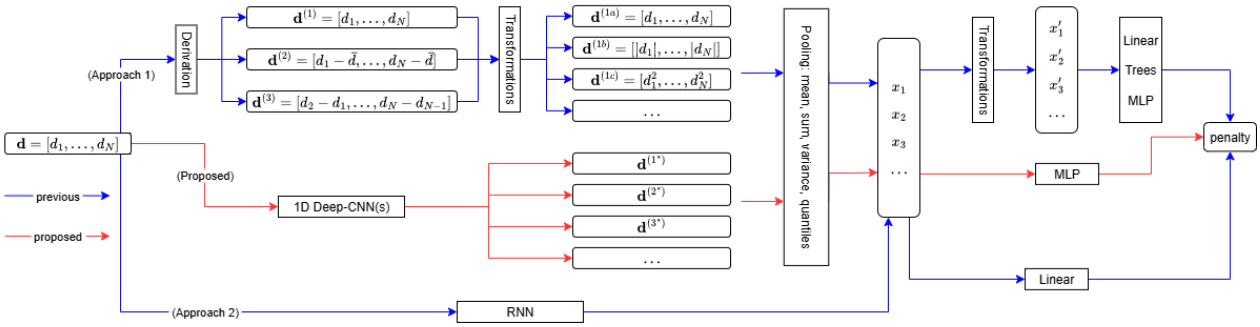


Figure 24: Diagram of the proposed model using 1D-CNN

regarded as state-of-the-art methods. These algorithms solve an optimization problem that balances data fidelity with a penalty parameter λ, which directly controls the trade-off between model complexity and goodness of fit. A larger λ results in fewer detected changepoints, while a smaller λ increases sensitivity to potential changes. Therefore, accurate prediction of λ is critical for achieving high detection performance.

Existing approaches for predicting λ include unsupervised criteria such as AIC-, BIC-, or related penalties [Schwarz, 1978, Akaike, 1974, Lavielle, 2005], as well as supervised models such as linear regression [Rigaill et al., 2013] and tree-based methods [Drouin et al., 2017, Barnwal et al., 2022, Nguyen and Hocking, 2025c]. However, these approaches depend heavily on manually engineered features extracted from the input sequence. Since the space of possible features is effectively unbounded, manual feature design may include irrelevant characteristics while overlooking informative ones, thereby limiting predictive performance.

In earlier versions of this work, we incorporated a recurrent neural network (RNN) to automatically learn sequence representations for λ prediction, alleviating the need for manual feature engineering. While the RNN-based model achieved strong predictive performance, it suffered from long training times due to the sequential nature of recurrent computation, which restricts parallelization and increases computational cost, especially for long sequences.

In this study, we propose a more computationally efficient alternative based on an ensemble

of one-dimensional convolutional neural networks (1D-CNNs) combined with a multilayer perceptron (MLP). The 1D-CNN components extract hierarchical temporal features from the sequence in parallel, and the MLP aggregates these learned representations to predict the penalty parameter λ. The proposed architecture achieves performance comparable to the RNN-based model in terms of OPART accuracy, while significantly reducing training time. Importantly, the proposed model is not a heuristic neural network–based changepoint detector; rather, it learns to predict the penalty parameter for a classical DP-based changepoint algorithm grounded in maximum likelihood inference.

7.2 Heuristic Partitioning for circular data

Circular (directional or angular) data naturally arise in numerous scientific fields. Typical applications include wind direction monitoring Harvey et al. [2023], wave direction analysis Jona-Lasinio et al. [2012], and molecular biology Sowa et al. [2005]. In particular, protein backbone dihedral angles provide a fundamental example of circular measurements. The protein dataset and the associated avocado plots considered in this work are introduced later.

Let $I := [-\pi, \pi)$. The geodesic (arc-length) distance on I is defined, for all $\theta, \theta' \in I$, by

$$d(\theta, \theta') = \min\left(|\theta - \theta'|,\ 2\pi - |\theta - \theta'|\right).$$

This geometry makes even basic statistical tasks nontrivial. For instance, computing a simple average on the circle generally admits no closed-form expression and requires dedicated optimization algorithms Hotz and Huckemann [2015], Arnaudon and Miclo [2016], Cazals et al. [2021], McKilliam et al. [2012]. The distinction between intrinsic and extrinsic means has been extensively studied Hotz [2013], McKilliam et al. [2012]. Statistical inference for circular data dates back to the 1960s Watson [1966], Watson and Beran [1967], and a large body of work now addresses distributions and parameter estimation on the circle Coeurjolly and Bihan [2012], Ameijeiras-Alonso and Ley [2022], Jona-Lasinio et al. [2012]. In particular, the sine-skewed toroidal distribution is a

classical model for dihedral angle distributions Ameijeiras-Alonso and Ley [2022].

Modern developments include regression methods Nguyen et al. [2023] and principal component analysis on the circle García-Portugués and Prieto-Tirado [2023]. Signal processing techniques have also been extended to circular data, including autoregressive processes Fisher and Lee [1994, 1983], Harvey et al. [2023], hidden Markov models Holzmann et al. [2006], Beh et al. [2014], Lagona et al. [2015], Lagona and Mingione [2025], Gaussian processes Jona-Lasinio et al. [2012], median filtering Storath and Weinmann [2018], signal approximation Condat [2022], and change-point detection Seifert et al. [2013], Storath et al. [2017]. Comprehensive reviews are available in Lee [2010], Pewsey and García-Portugués [2021].

Change-point detection is particularly important in practice. In protein analysis, structural regimes correspond to distinct conformational states, making segmentation methods especially relevant. However, existing approaches are typically restricted to the univariate setting and become computationally impractical for very long signals (e.g., several hundred thousand samples). Moreover, classical dynamic programming methods cannot be directly applied because they require repeated updates of a mean parameter, which is intractable in the angular setting due to the non-convexity of the Fréchet functional McKilliam et al. [2012], Cazals et al. [2021].

Signal model We observe a sequence of I-valued random variables $Y_1, \dots, Y_T$ satisfying

$$Y_t = [\theta^\star(t/T) + \epsilon_t \bmod 2\pi] - \pi, \tag{16}$$

where $\theta^\star : [0,1) \to I$ is a piecewise constant deterministic function and $(\epsilon_t)_t \subset I$ is a noise process.

There exist $K^\star$ change-points $0 < \tau_1^\star < \cdots < \tau_{K^\star}^\star < 1$ and $K^\star + 1$ levels $m_k^\star \in I$ such that, for all $\tau \in [0,1)$,

$$\theta^\star(\tau) = \sum_{k=1}^{K^\star+1} m_k^\star \, \mathbf{1}\big(\tau_{k-1}^\star \le \tau < \tau_k^\star\big) \,,$$

with the convention $\tau_0^\star := 0$ and $\tau_{K^\star+1}^\star := 1$.

The objective of change-point detection is to estimate the unknown number of change-points

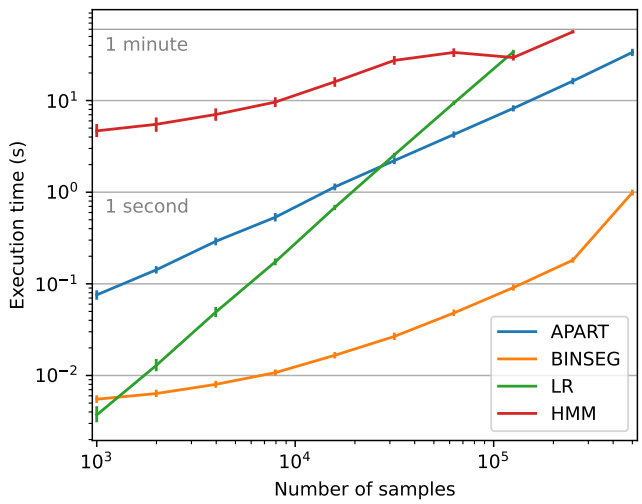


Figure 25: Operation time of Changepoint Detection models

$K^\star$ and their locations $\tau_k^\star$.

Change-point formulation Given an angular signal $Y \in I^T$ and a penalty parameter $\beta > 0$, change-point detection can be formulated as

$$\min_{\boldsymbol{\theta} \in I^T} \left[\sum_{t=1}^{T} d^2(Y_t, \theta_t) + \beta \sum_{t=1}^{T-1} \mathbf{1}(\theta_t \neq \theta_{t+1}) \right]. \tag{17}$$

Classical dynamic programming approaches (e.g., Harvey et al. [2023], Lee [2010], Jona-Lasinio et al. [2012], Lagona et al. [2015], Pewsey and García-Portugués [2021], Beh et al. [2014]) are not directly applicable because they require updating a segment-wise mean parameter. In the circular setting, this update involves minimizing a non-convex Fréchet functional, which is computationally demanding and lacks closed-form solutions.

Alternative procedures include binary segmentation with hypothesis testing under Fisher–von Mises distribution shifts Jammalamadaka and Sengupta [2001] and hidden Markov models Michelot et al. [2016]. However, these approaches are mainly restricted to univariate signals and become computationally impractical for very long sequences.

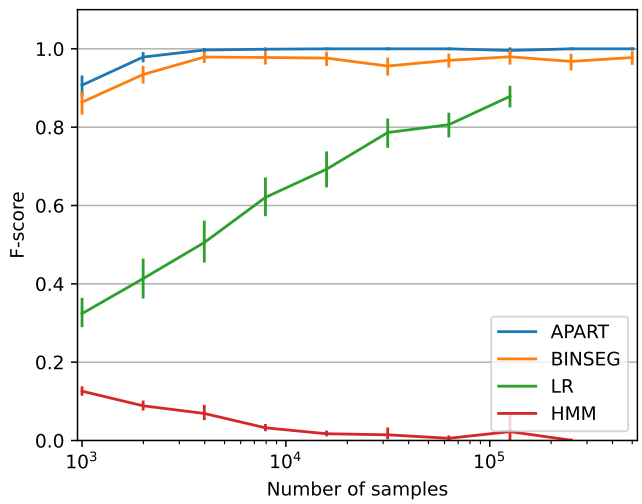


Figure 26: F-score of Changepoint Detection models

To address these limitations, we restrict θ_t to a finite set $\Theta := \{\theta^{(1)}, \ldots, \theta^{(M)}\} \subset I$. The set Θ does not need to contain the true levels $m_k^\star$, but only approximate them sufficiently well. Problem (17) then becomes

$$\min_{(\theta_t)_t \in \Theta^T} \left[\sum_{t=1}^{T} d^2(Y_t, \theta_t) + \beta \sum_{t=1}^{T-1} \mathbf{1}(\theta_t \neq \theta_{t+1}) \right]. \tag{18}$$

This discrete formulation admits an efficient dynamic programming solution with time complexity $\mathcal{O}(|\Theta|T)$ and space complexity $\mathcal{O}(|\Theta|T)$.

Contributions The main contributions of this work are as follows. First, we establish finite-sample recovery guarantees for the proposed method. Second, we prove that the estimator achieves the minimax optimal rate. Third, we introduce an improved initialization procedure that enhances practical performance. Fourth, we provide an open-source implementation available in multiple programming languages. Finally, we demonstrate the practical relevance of our approach on real protein dihedral angle data.

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